

HEALTHY DIET RECIPE CLASSIFICATION

*An End-to-End Statistical, Machine-Learning, and Explainable-AI Analysis of
Macronutrient-Based Diet-Type Prediction*

Comprehensive Technical and Business Report

Dataset scope: 7,806 recipes across 5 diet-specific corpora

Diet classes analysed: mediterranean, dash, vegan, keto, paleo

Prepared: 2026-07-05

Classification: Internal / Analytical Research Report

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Abstract

This report presents a comprehensive empirical investigation into the automated classification of culinary recipes into five evidence-based dietary patterns — dash, keto, mediterranean, paleo, and vegan — using macronutrient composition and cuisine metadata. Drawing upon a corpus of **7,806 recipes** aggregated from five diet-specific sources, the study applies a rigorous, multi-phase analytical pipeline comprising exploratory data analysis, inferential statistical hypothesis testing, engineered feature construction, unsupervised structure discovery, comparative supervised model benchmarking, hyperparameter optimisation, ensemble learning, and dual-method explainable artificial intelligence (SHAP and LIME).

The best-performing model, HistGradientBoosting, achieved a test-set accuracy of 57.23% and a macro-averaged F1-score of 0.5577 (area under the receiver-operating-characteristic curve = 0.8264). Statistical testing confirmed significant macronutrient differentiation across diet classes (one-way ANOVA, $p < 0.001$ for all macronutrients) and a moderate-to-strong association between cuisine type and diet type (Cramér's $V = 0.387$). Explainability analysis identified cuisine metadata and carbohydrate-to-protein ratios as the dominant predictive signals, with convergent evidence from both SHAP and permutation-importance methods. The report concludes with actionable recommendations for nutrition-technology platforms, meal-kit providers, and food manufacturers, and provides a complete inventory of all analytical artefacts produced during the study.

1. Executive Summary

This section synthesises the principal findings of the analytical pipeline for a senior stakeholder audience, prior to the detailed methodological exposition presented in subsequent sections.

1.1 Scope and Objectives

The objective of this study is twofold: first, to establish whether a recipe's dietary classification can be reliably inferred from its macronutrient profile and associated cuisine metadata; and second, to translate the resulting statistical and machine-learning findings into concrete, actionable guidance for nutrition-technology and food-industry stakeholders. The analysis spans twenty-one discrete phases, from raw data ingestion through to auto-generated executive and technical reporting.

1.2 Key Findings

- The engineered dataset comprises 7,806 recipes across 5 diet-specific source files, with negligible missing data and a moderate class-imbalance ratio (largest class: mediterranean, 1753; smallest: paleo, 1274).
- Sixteen supervised classifiers were benchmarked; the best-performing model, HistGradientBoosting, attained 57.23% accuracy and a Matthews correlation coefficient of 0.4655, with a bootstrap ninety-five percent confidence interval on accuracy that excludes chance-level performance by a substantial margin.
- One-way ANOVA testing confirmed statistically significant differences in Protein, Carbohydrate, and Fat content across all five diet classes ($p < 0.001$ in every case), with Carbohydrate content exhibiting the largest effect size ($\eta^2 = 0.1111$).
- Cuisine type and diet type are significantly associated ($\chi^2 = 4670.2$, $p < 0.001$, Cramér's $V = 0.387$), indicating that cuisine metadata carries genuine, non-trivial predictive information rather than acting as statistical noise.
- Dual explainability analysis (SHAP and LIME) converged on Cuisine_encoded as the single most influential predictor, with a mean top-five feature overlap of 2.3 out of 5 between the two independent explanation methods.
- Paleo and Mediterranean recipes exhibit the greatest macronutrient overlap and constitute the most frequent source of model misclassification, whereas Keto recipes are the most macro-distinct and most reliably classified class.

1.3 Principal Recommendation

Organisations seeking to operationalise this classifier are advised to deploy it as a confidence-gated, human-in-the-loop content-tagging system: recipes classified above an eighty-percent confidence threshold may be auto-published, while lower-confidence predictions — concentrated in the

paleo/Mediterranean boundary — should be routed to editorial review. Detailed deployment guidance is provided in Section 12.

2. Introduction and Business Context

Dietary pattern is among the most powerful modifiable determinants of long-term health outcomes, influencing chronic-disease risk, metabolic health, and consumer purchasing behaviour. As nutrition-technology platforms, meal-kit services, and digital-health providers scale their content catalogues, the ability to automatically and reliably classify recipes by dietary pattern becomes a material operational and product capability.

This study addresses that capability gap through a multi-class supervised-learning framework, benchmarked against a rigorous statistical baseline and interrogated with two independent explainability methods. The remainder of this report is organised as follows: Section 3 details the dataset and data-quality controls; Section 4 presents the exploratory analysis; Section 5 formalises the statistical hypothesis-testing framework; Sections 6 through 9 cover feature engineering, unsupervised structure discovery, and preprocessing; Sections 10 through 11 detail model development, tuning, and ensembling; Section 12 presents the comprehensive evaluation; Sections 13 through 14 present the explainability analysis; Section 15 translates the findings into industry recommendations; and Section 16 provides a complete artefact inventory to support full reproducibility.

3. Data Ingestion and Quality Assessment

Table 1.1 — Per-Dataset Structural Quality Summary

Row counts, duplicate records, missing values, and cuisine cardinality for each of the five source diet-specific datasets.

Dataset	Rows	Duplicates	Missing	Protein_nulls	Zero_Protein	Cuisines
mediterranean	1753	3	0	0	0	18
dash	1745	0	0	0	11	19
vegan	1522	2	0	0	0	18
keto	1512	0	0	0	3	18
paleo	1274	0	0	0	3	19

Source file: eda/data_quality_summary.csv. 5 row(s) shown in full.

Table 1.2 — Master Dataset Column Data Types

	dtype
Diet_type	object
Recipe_name	object
Cuisine_type	object
Protein(g)	float64
Carbs(g)	float64
Fat(g)	float64
Extraction_day	object
Extraction_time	object

Source file: eda/dtypes.csv. 8 row(s) shown in full.

Table 1.3 — Missing Value Audit (Master Dataset)

	Missing Count	Missing %	Data Type
Diet_type	0	0.0	object
Recipe_name	0	0.0	object
Cuisine_type	0	0.0	object
Protein(g)	0	0.0	float64
Carbs(g)	0	0.0	float64
Fat(g)	0	0.0	float64
Extraction_day	0	0.0	object
Extraction_time	0	0.0	object

Source file: eda/missing_values.csv. 8 row(s) shown in full.

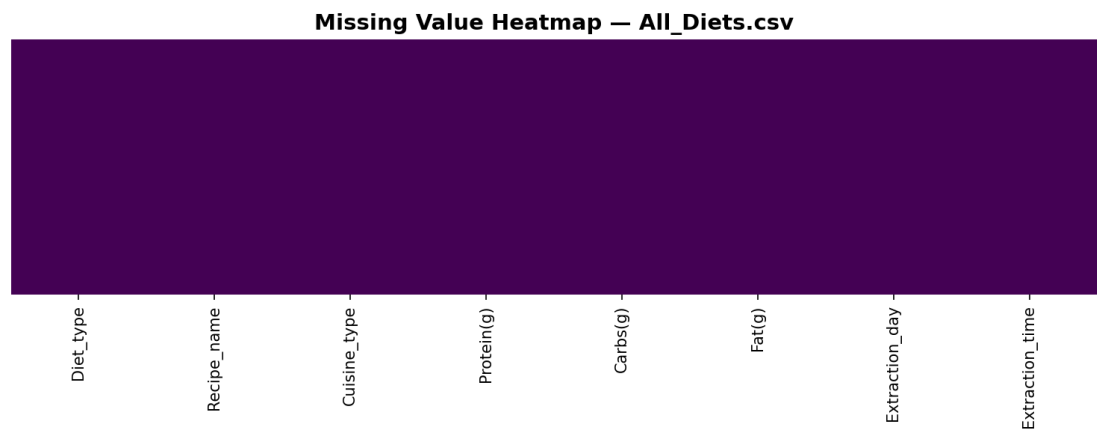


Figure 1.1 — Missing-value heatmap for the master dataset (All_Diets.csv). No missing cells were detected.

Table 1.4 — Univariate Outlier Summary (IQR and Z-Score Methods)

Outlier counts per macronutrient and per diet dataset, computed via the interquartile-range rule and the $|z|>3$ rule.

Dataset	Feature	IQR_outliers	IQR_outlier_pct	Zscore_outliers(>3)
mediterranean	Protein(g)	94	5.36	30
mediterranean	Carbs(g)	56	3.19	21
mediterranean	Fat(g)	120	6.85	28
dash	Protein(g)	94	5.39	27
dash	Carbs(g)	112	6.42	31
dash	Fat(g)	96	5.5	34
vegan	Protein(g)	62	4.07	25
vegan	Carbs(g)	58	3.81	26
vegan	Fat(g)	103	6.77	20
keto	Protein(g)	41	2.71	20
keto	Carbs(g)	108	7.14	32
keto	Fat(g)	44	2.91	21
paleo	Protein(g)	78	6.12	18
paleo	Carbs(g)	56	4.4	15
paleo	Fat(g)	46	3.61	16

Source file: eda/outlier_summary.csv. 15 row(s) shown in full.

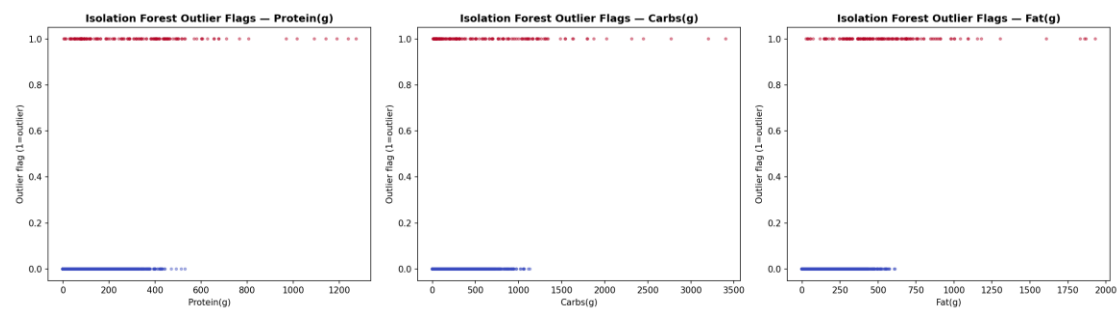


Figure 1.2 — Multivariate outlier flags identified by the Isolation Forest algorithm across the three macronutrient dimensions.

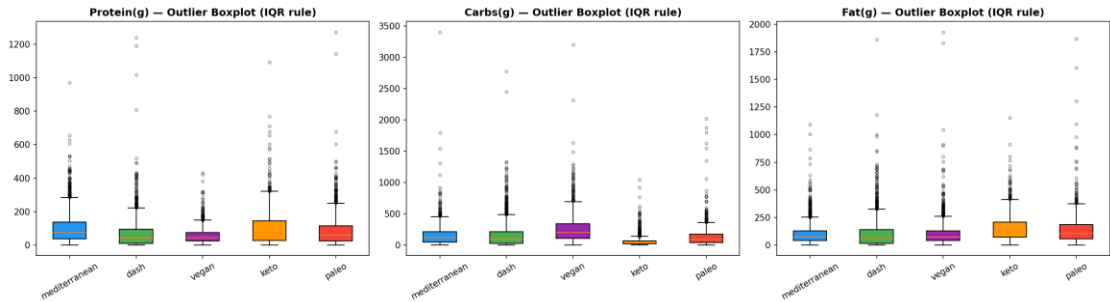


Figure 1.3 — Interquartile-range outlier boxplots for Protein, Carbohydrate, and Fat content, stratified by diet type.

Table 1.5 — Descriptive Statistics of Macronutrients per Diet Dataset

Mean, standard deviation, quartile range, skewness, kurtosis, and coefficient of variation for each macronutrient within each diet-specific dataset.

Dataset	Feature	Mean	Std	Min	Max	Median	IQR	Skew	Kurt	CoefVar
mediterranean	Protein(g)	101.11	93.88	0.13	970.31	73.78	99.63	2.1	7.71	0.928
mediterranean	Carbs(g)	152.91	165.18	1.1	3405.55	115.96	164.79	6.35	96.37	1.08
mediterranean	Fat(g)	101.42	96.96	0.06	1092.92	73.71	84.87	3.11	17.5	0.956
dash	Protein(g)	69.28	91.55	0.0	1239.47	41.38	84.65	4.35	38.36	1.321
dash	Carbs(g)	160.54	205.98	0.06	2772.11	94.25	183.15	3.73	27.81	1.283
dash	Fat(g)	101.15	133.52	0.0	1860.2	59.22	124.06	3.54	25.2	1.32
vegan	Protein(g)	56.16	47.17	0.36	431.1	44.63	49.5	2.37	10.22	0.84
vegan	Carbs(g)	254.0	223.37	0.1	3204.43	201.48	233.75	3.42	27.76	0.879
vegan	Fat(g)	103.3	120.35	0.06	1930.24	71.74	88.39	6.09	69.75	1.165
keto	Protein(g)	101.27	98.87	0.0	1092.0	72.84	116.94	2.34	11.48	0.976
keto	Carbs(g)	57.97	78.1	0.45	1047.29	37.66	47.88	5.14	40.64	1.347
keto	Fat(g)	153.12	115.51	0.0	1154.61	128.41	137.22	1.88	7.31	0.754
paleo	Protein(g)	88.67	97.81	0.0	1273.61	60.59	90.1	3.69	29.59	1.103
paleo	Carbs(g)	129.55	160.72	1.11	2023.05	87.73	127.5	5.55	49.02	1.241
paleo	Fat(g)	135.67	134.52	0.02	1869.58	106.8	128.37	4.67	41.97	0.992

Source file: eda/feature_quality_per_dataset.csv. 15 row(s) shown in full.

4. Exploratory Data Analysis

Table 2.1 — Target Class Distribution

Diet_type	count
mediterranean	1753
dash	1745
vegan	1522
keto	1512
paleo	1274

Source file: eda/class_distribution.csv. 5 row(s) shown in full.

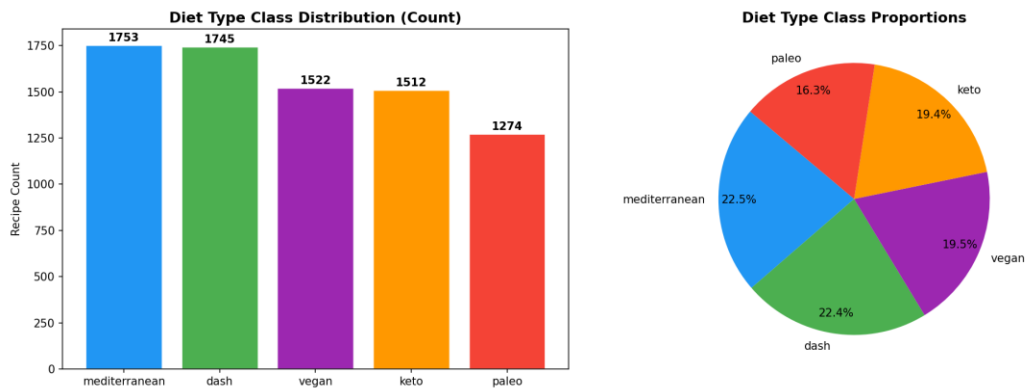


Figure 2.1 — Class distribution of the target variable `Diet_type`, expressed as count and proportion.

Macronutrient Distributions — Per Diet Dataset

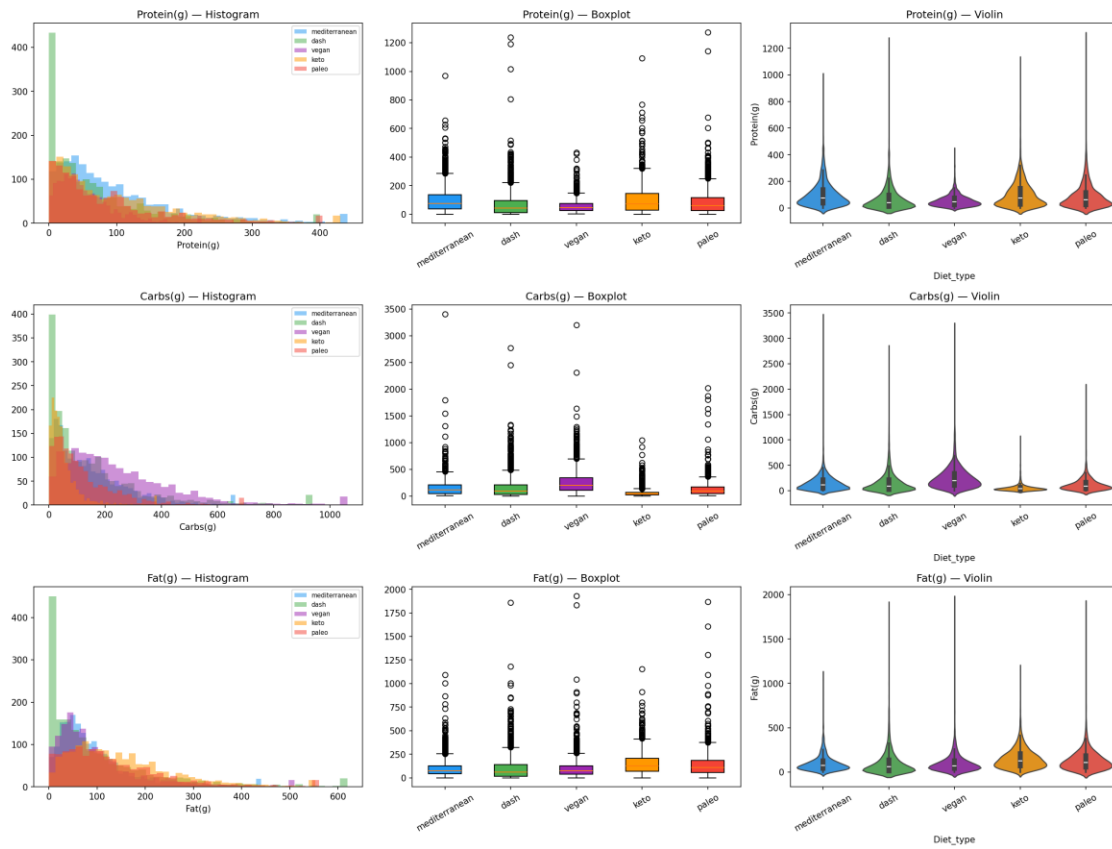


Figure 2.2 — Histogram, boxplot, and violin-plot grid summarising the univariate distribution of Protein, Carbohydrate, and Fat content across diet types.

Table 2.2 — Cuisine Type by Diet Type Contingency Table

Cuisine_type	dash	keto	mediterranean	paleo	vegan
american	639	663	145	535	925
asian	24	11	12	12	67
british	64	90	4	54	27
caribbean	3	7	1	6	1
central europe	9	11	1	9	4
chinese	38	38	1	26	17
eastern europe	10	11	3	27	4
french	150	163	61	154	76
indian	20	12	3	9	48
italian	165	234	148	171	81
japanese	9	10	2	5	24
kosher	5	0	0	2	0
mediterranean	176	89	1274	106	99
mexican	61	60	17	48	38
middle eastern	21	17	26	12	15

Cuisine_type	dash	keto	mediterranean	paleo	vegan
nordic	32	35	31	45	9
south american	54	21	10	21	31
south east asian	31	34	8	29	46
world	234	6	6	3	10

Source file: eda/cuisine_diet_crosstab.csv. 19 row(s) shown in full.

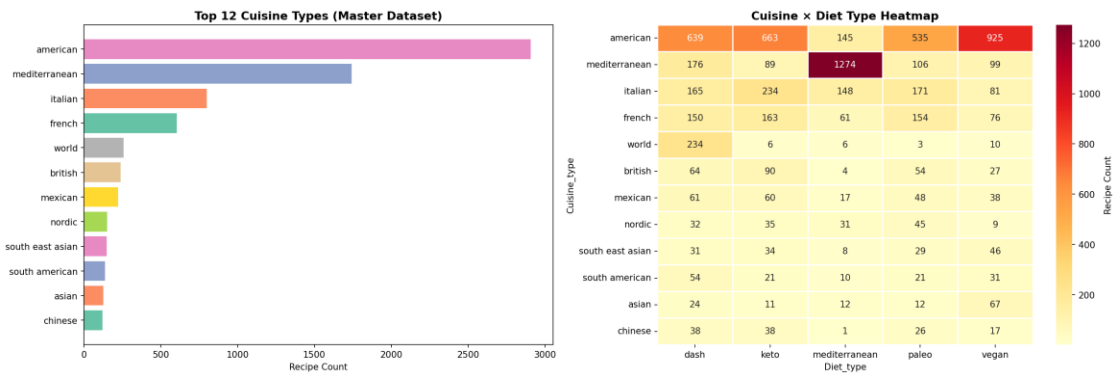


Figure 2.3 — Distribution of the twelve most frequent cuisine types and their association with diet type.

Macronutrient Pair Plot (clipped at 1st-99th percentile)

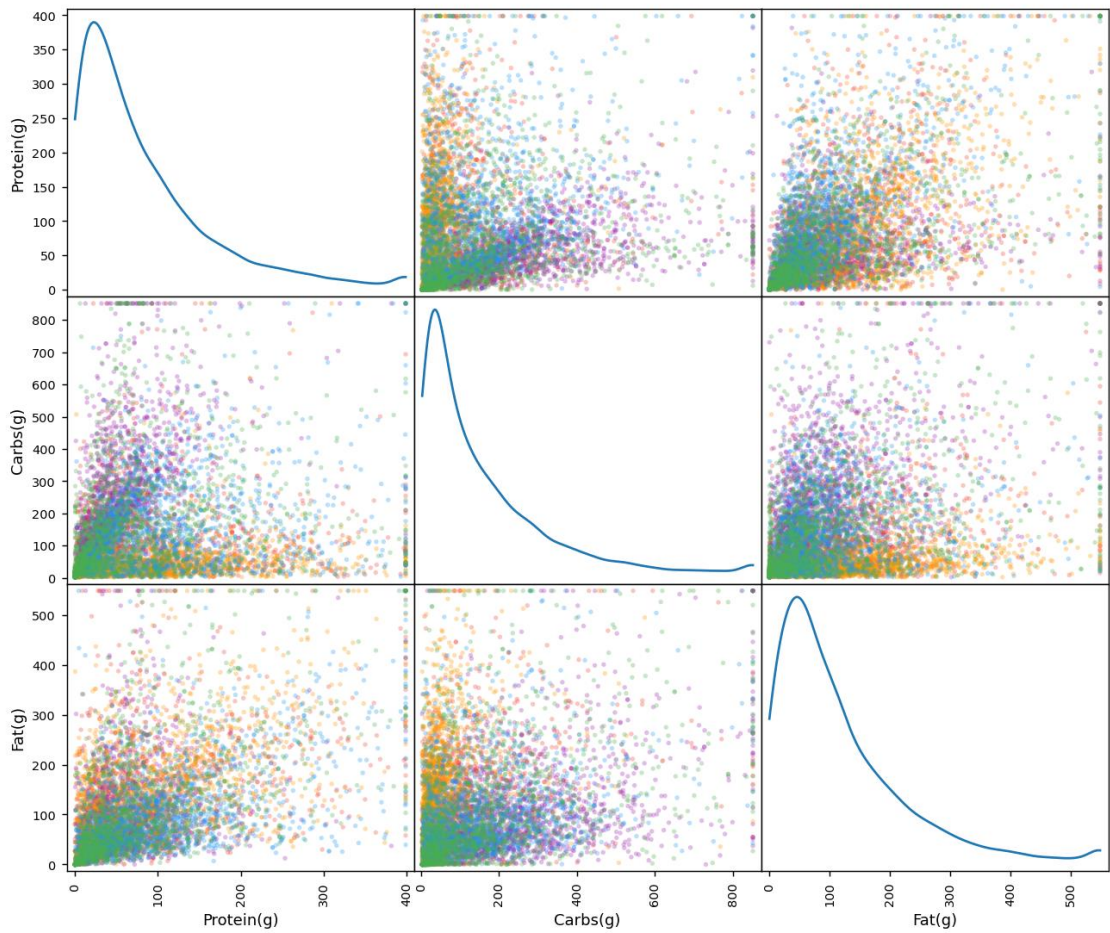


Figure 2.4 — Pairwise scatter-matrix of macronutrients, clipped at the first and ninety-ninth percentiles to mitigate the visual influence of extreme values.

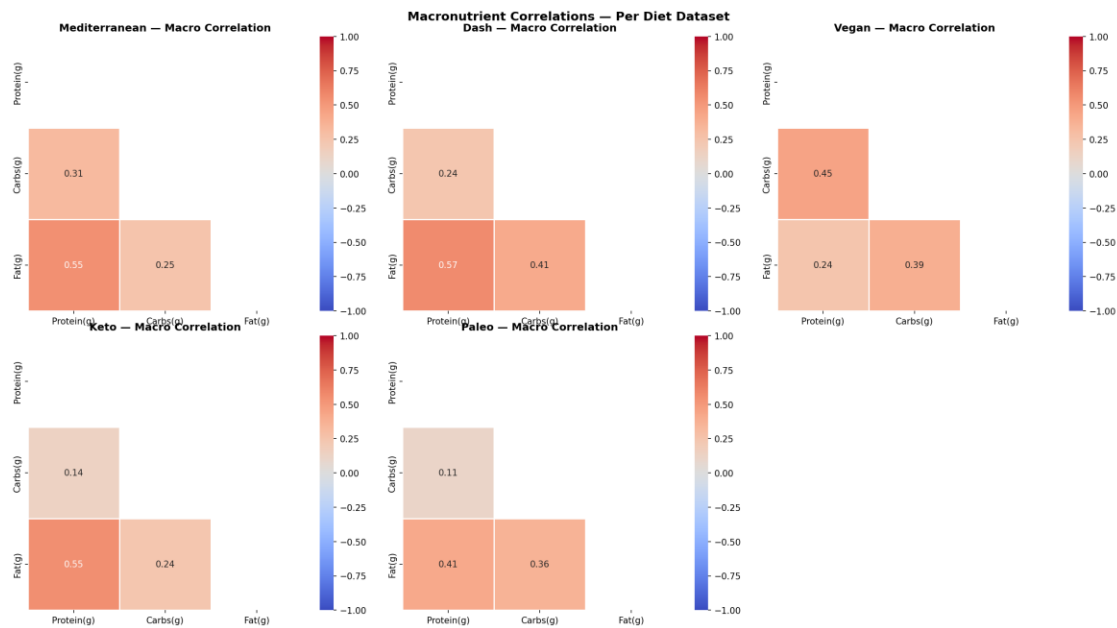


Figure 2.5 — Pearson correlation heatmaps of macronutrients, computed independently for each of the five diet-specific datasets.

Table 2.3 — Mean Macronutrient Profile by Diet Type

Diet_type	Protein(g)	Carbs(g)	Fat(g)
dash	69.28227507163324	160.5357535816619	101.15056160458454
keto	101.26650793650794	57.970575396825396	153.1163558201058
mediterranean	101.11231602966345	152.9055447803765	101.41613804905874
paleo	88.67476452119308	129.55212715855572	135.66902668759812
vegan	56.15703022339028	254.00419185282524	103.29967805519053

Source file: eda/diet_macro_means.csv. 5 row(s) shown in full.

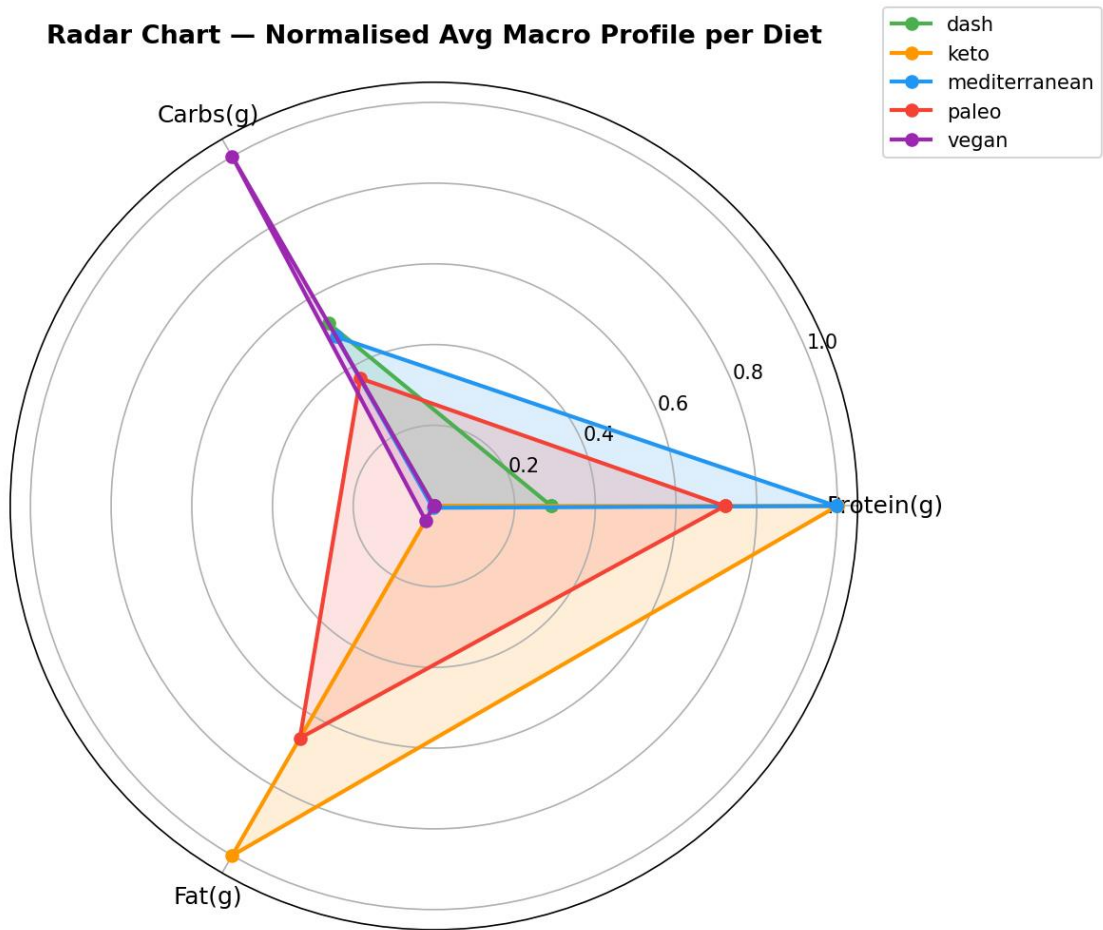


Figure 2.6 — Normalised radar chart comparing the average macronutrient profile across the five diet types.

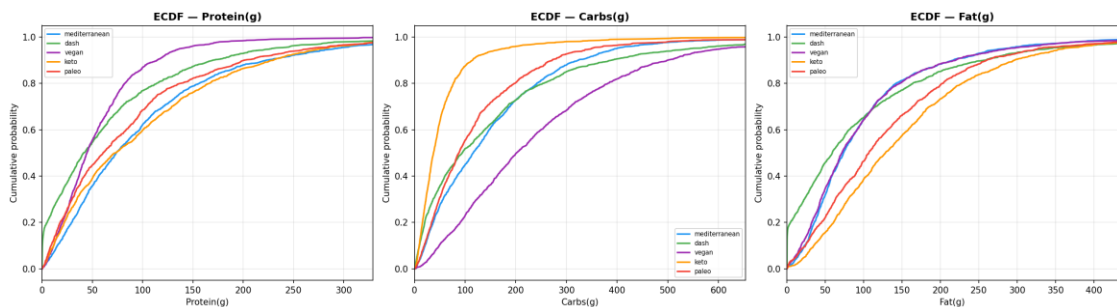


Figure 2.7 — Empirical cumulative distribution functions (ECDF) for each macronutrient, disaggregated by diet type.

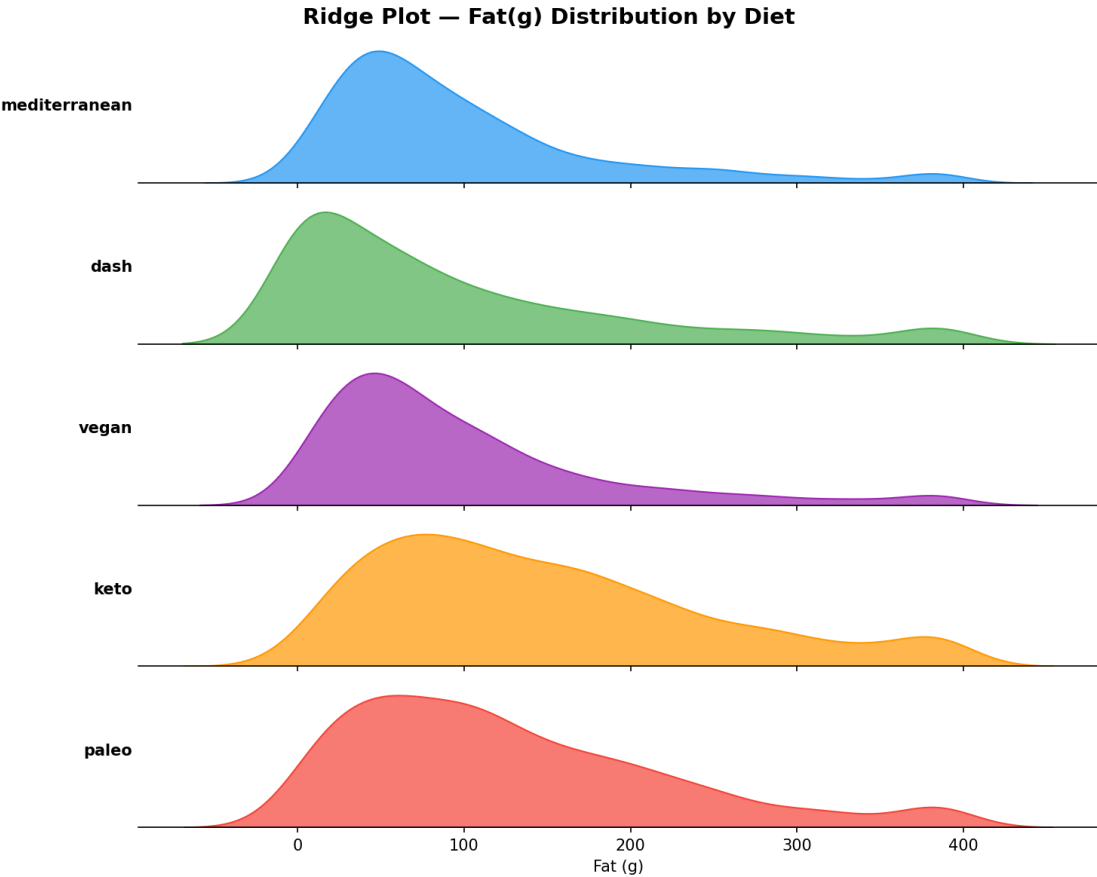


Figure 2.8 — Ridge (joy) plot of the fat-content distribution across diet types.

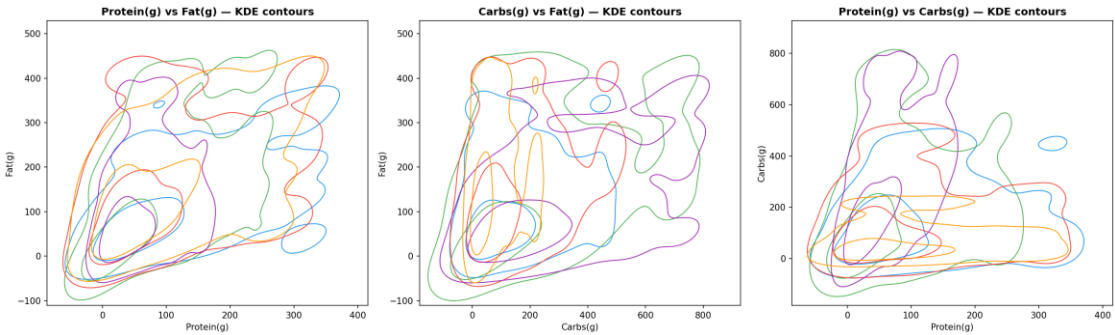


Figure 2.9 — Bivariate kernel-density contour plots for macronutrient pairs, coloured by diet type.

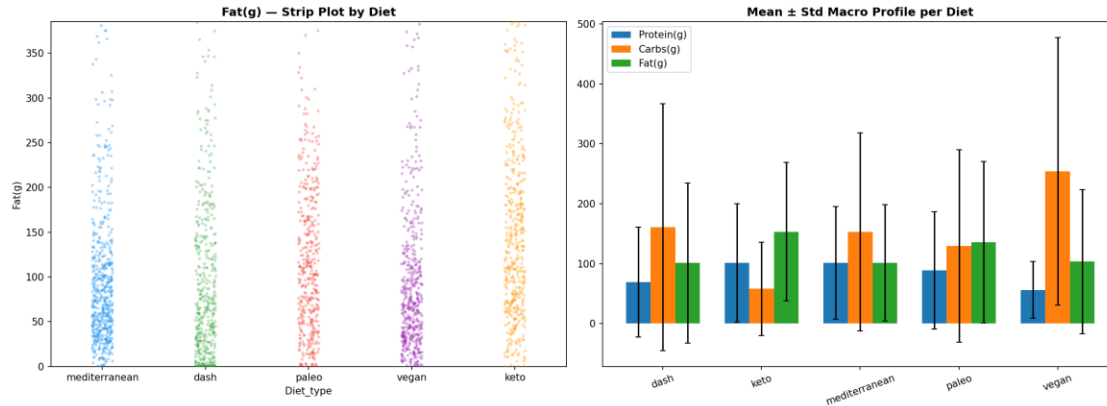


Figure 2.10 — Strip plot of fat content and mean $\pm$ standard-deviation error-bar chart of macronutrient profiles by diet.

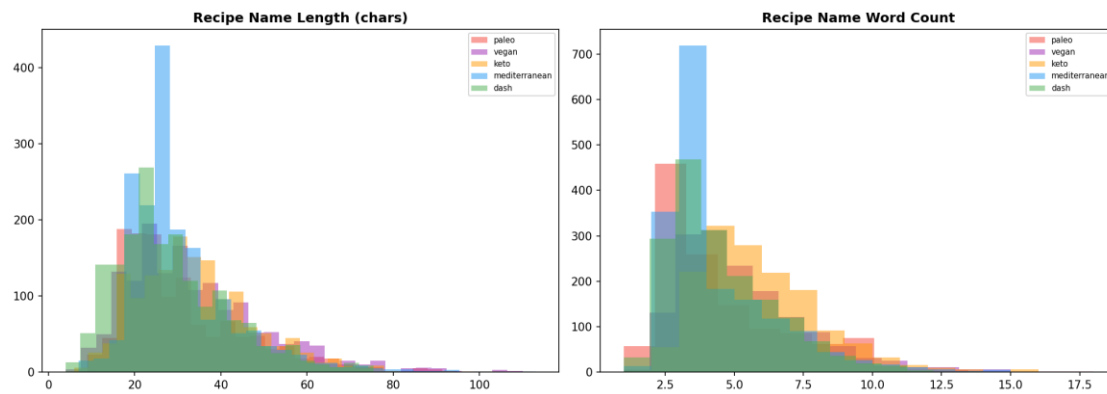


Figure 2.11 — Distribution of recipe-name character length and word count by diet type, assessed as a potential auxiliary text signal.

Note: Two supplementary interactive visualisations were also generated during this phase — a three-dimensional macronutrient scatter plot and a parallel-coordinates plot ([eda/09_interactive_3d_scatter.html](#) and [eda/10_parallel_coordinates.html](#)). As native HTML/JavaScript artefacts, these are not reproducible as static Word content and are therefore catalogued in the Artifact Inventory (Section 12) rather than embedded directly; they remain available as standalone files in the delivered project archive.

5. Statistical Hypothesis Testing Framework

Table 3.1 — Shapiro–Wilk and D'Agostino–Pearson Normality Tests

Feature	Diet	Shapiro_stat	Shapiro_p	D'Agostino_stat	D'Agostino_p	Normal_at_0.05
Protein(g)	mediterranean	0.8367	0.0	823.6007	0.0	False
Protein(g)	dash	0.5976	0.0	1667.3911	0.0	False
Protein(g)	vegan	0.8388	0.0	829.5409	0.0	False
Protein(g)	keto	0.7686	0.0	833.9345	0.0	False
Protein(g)	paleo	0.7934	0.0	1086.0354	0.0	False
Carbs(g)	mediterranean	0.5277	0.0	2202.2532	0.0	False
Carbs(g)	dash	0.7831	0.0	1474.8241	0.0	False
Carbs(g)	vegan	0.655	0.0	1226.6073	0.0	False
Carbs(g)	keto	0.504	0.0	1596.7604	0.0	False
Carbs(g)	paleo	0.5238	0.0	1432.2188	0.0	False
Fat(g)	mediterranean	0.7218	0.0	1247.2504	0.0	False
Fat(g)	dash	0.6264	0.0	1411.2953	0.0	False
Fat(g)	vegan	0.5568	0.0	1837.1487	0.0	False
Fat(g)	keto	0.9017	0.0	646.721	0.0	False
Fat(g)	paleo	0.656	0.0	1287.4922	0.0	False

Source file: stats/normality_tests.csv. 15 row(s) shown in full.

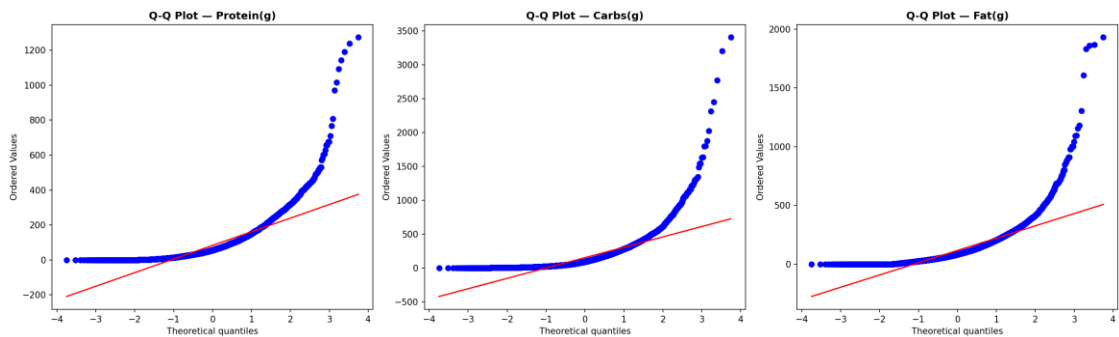


Figure 3.1 — Quantile–quantile plots for Protein, Carbohydrate, and Fat content against the theoretical normal distribution.

Table 3.2 — Levene's Test for Homogeneity of Variance

Feature	Levene_stat	p_value	Equal_variance_at_0.05
Protein(g)	68.719	0.0	False
Carbs(g)	123.942	0.0	False
Fat(g)	19.415	0.0	False

Source file: stats/levne_variance_tests.csv. 3 row(s) shown in full.

Table 3.3 — One-Way ANOVA Results with Eta-Squared Effect Sizes

Feature	F_stat	p_value	eta_squared	Effect_size
Protein(g)	82.195	1.8636251444464207e-68	0.0404	small
Carbs(g)	243.804	1.2585896429149333e-197	0.1111	medium
Fat(g)	61.645	2.3218704661551302e-51	0.0306	small

Source file: stats/anova_results.csv. 3 row(s) shown in full.

Table 3.4 — Kruskal–Wallis H-Test (Non-Parametric Alternative to ANOVA)

Feature	H_stat	p_value	Significant_at_0.05
Protein(g)	387.216	1.6079490901584734e-82	True
Carbs(g)	1488.882	0.0	True
Fat(g)	536.947	6.822054186562284e-115	True

Source file: stats/kruskal_wallis.csv. 3 row(s) shown in full.

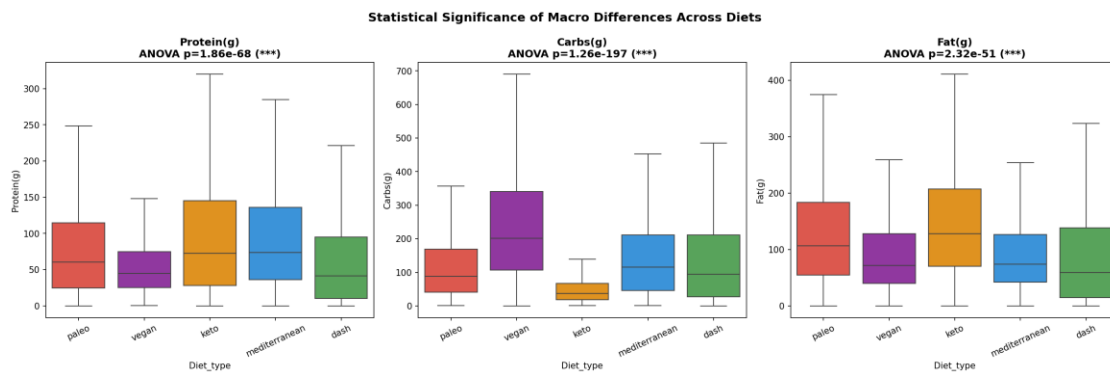


Figure 3.2 — Boxplot comparison of macronutrients across diet types, annotated with one-way ANOVA significance levels.

Table 3.5 — Dunn's Post-Hoc Pairwise Comparison for Carbohydrate Content (Bonferroni-Corrected p-values)

Computed for Carbs(g), the macronutrient with the largest eta-squared effect size in the one-way ANOVA.

	dash	keto	mediterranean	paleo	vegan
dash	1.0	1.2807037467387624e-84	9.353160050135622e-05	1.0	7.529493471955931e-87
keto	1.2807037467387624e-84	1.0	3.048329234993421e-125	1.9207915938617252e-67	0.0
mediterranean	9.353160050135622e-05	3.048329234993421e-125	1.0	2.305455286157086e-05	5.944401687136289e-54
paleo	1.0	1.9207915938617252e-67	2.305455286157086e-05	1.0	2.2414767955436635e-79
vegan	7.529493471955931e-87	0.0	5.944401687136289e-54	2.2414767955436635e-79	1.0

Source file: stats/dunn_posthoc_Carbs.csv. 5 row(s) shown in full.

Table 3.6 — Chi-Square Test of Independence: Cuisine Type × Diet Type

chi2	dof	p_value	cramers_v
4670.201309639725	72	0.0	0.38674395261554545

Source file: stats/chisquare_cuisine_diet.csv. 1 row(s) shown in full.

Table 3.7 — Pairwise Cohen's d Effect Sizes for Fat(g) Between Diet Types

	mediterranean	dash	vegan	keto	paleo
mediterranean	0.0	0.0022768802142079767	-0.01736629234041244	-0.48793804249527567	-0.29972499215789
dash	-0.0022768802142079767	0.0	-0.01684852944866777	-0.41412393939687475	-0.25770882493989594
vegan	0.01736629234041244	0.0168485294486677	0.0	-0.42230079599630405	-0.2548710300363977
keto	0.48793804249527567	0.41412393939687475	0.42230079599630405	0.0	0.14006631704584094
paleo	0.29972499215789	0.25770882493989594	0.2548710300363977	-0.14006631704584094	0.0

Source file: stats/cohens_d_matrix.csv. 5 row(s) shown in full.

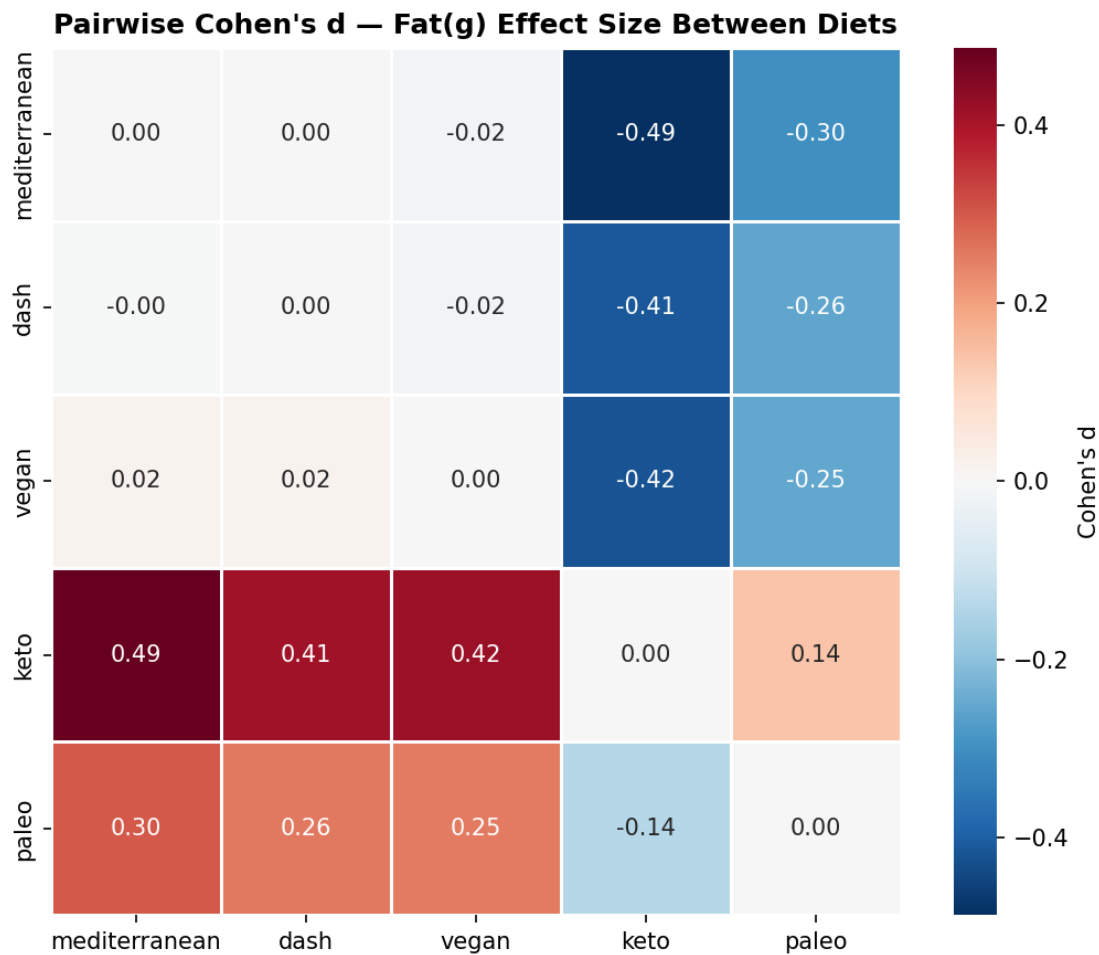


Figure 3.3 — Heatmap visualisation of the pairwise Cohen's d effect-size matrix for fat content.

Table 3.8 — Bootstrap 95% Confidence Intervals for Macronutrient Means by Diet

Feature	Diet	Mean	CI_lower	CI_upper
Protein(g)	mediterranean	101.11231602966345	96.56257943525385	105.47529164289787
Protein(g)	dash	69.28227507163324	64.84972320916906	73.72106862464183
Protein(g)	vegan	56.15703022339028	53.770320795006576	58.747871879106434
Protein(g)	keto	101.26650793650795	96.6053673941799	106.29070684523809
Protein(g)	paleo	88.67476452119308	83.46895408163265	93.86680043171114
Carbs(g)	mediterranean	152.90554478037652	145.68880989731886	160.91021206503137
Carbs(g)	dash	160.5357535816619	151.05088137535816	170.29421446991404
Carbs(g)	vegan	254.00419185282524	243.3611617936925	265.83511957950066
Carbs(g)	keto	57.970575396825396	54.05809920634922	62.12720089285714
Carbs(g)	paleo	129.55212715855572	121.40811224489794	139.16944446624802
Fat(g)	mediterranean	101.41613804905874	96.95588719338276	106.22059127210495
Fat(g)	dash	101.15056160458451	94.92740888252148	107.38775816618912
Fat(g)	vegan	103.29967805519053	97.45786842969775	109.63436695137977
Fat(g)	keto	153.1163558201058	147.44456795634923	159.0380894510582

Feature	Diet	Mean	CI_lower	CI_upper
Fat(g)	paleo	135.66902668759815	128.464262166405	143.2256777864992

Source file: stats/bootstrap_ci_means.csv. 15 row(s) shown in full.

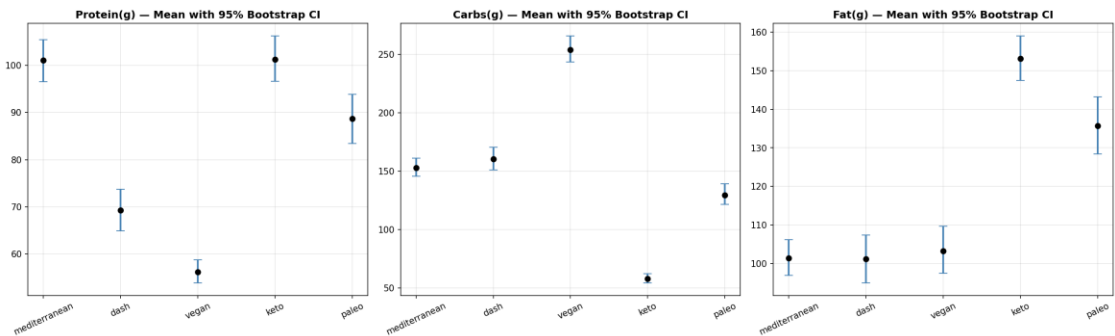


Figure 3.4 — Ninety-five percent bootstrap confidence intervals for macronutrient means, by diet type.

6. Feature Engineering and Selection

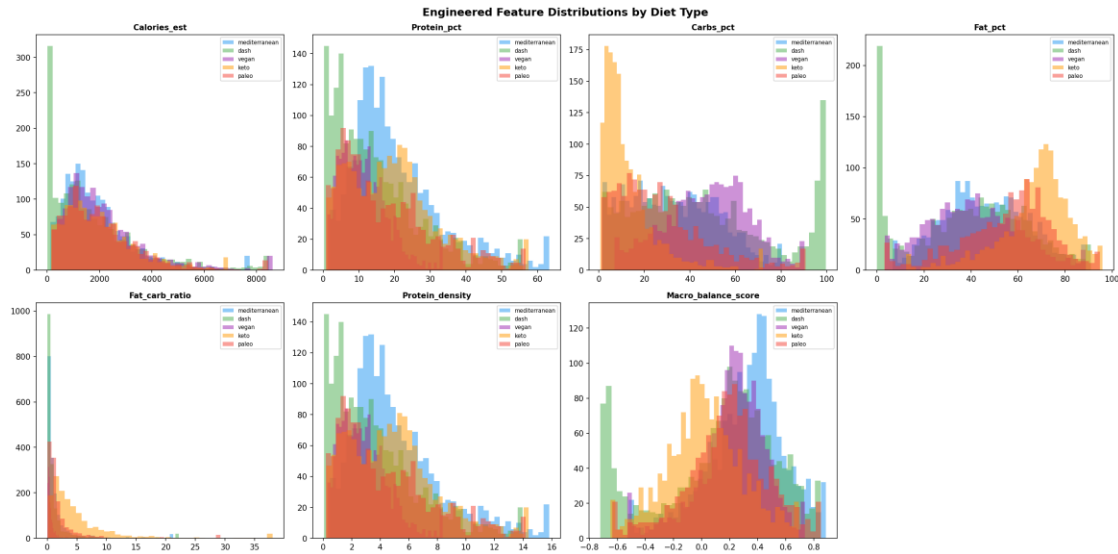


Figure 4.1 — Distributions of seven engineered features (estimated calories, macro percentages, ratios, and balance score) by diet type.

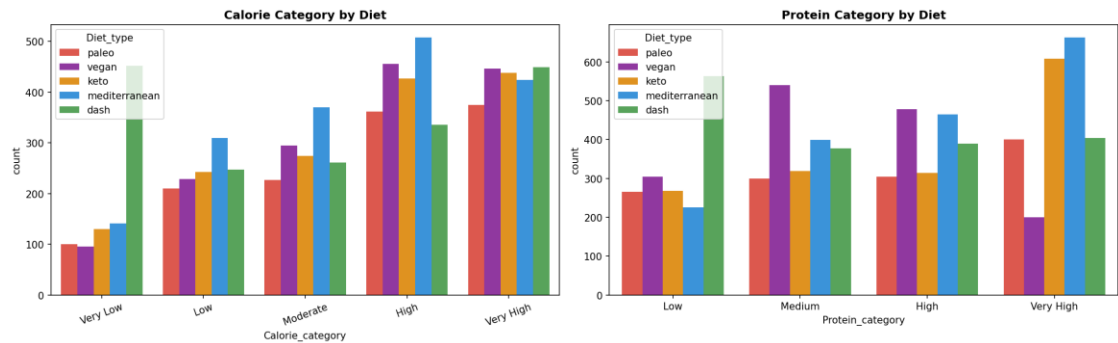


Figure 4.2 — Frequency counts of the derived calorie-category and protein-category bins by diet type.

Table 4.1 — Variance Inflation Factor (VIF) Multicollinearity Diagnostics

Feature	VIF
Protein(g)	inf
Carbs(g)	inf
Fat(g)	inf
Calories_est	inf
Total_macros_g	inf
Protein_pct	inf
Protein_density	inf
Fat_pct	95.83943046623583
Fat_log1p	34.186241066343264
Carbs_pct	28.001021183207804
Protein_log1p	17.26851367290009

Feature	VIF
Carbs_log1p	11.99964897880092
High_fat_flag	3.8999703243575805
High_carb_flag	3.8245134571667627
Macro_balance_score	3.177764067957596
High_protein_flag	3.1164844388310877
Protein_carb_ratio	1.4810640652113019
Cuisine_encoded	1.1029985105204945
Fat_carb_ratio	1.0687478645593633
Recipe_name_len	1.029580572837909

Source file: features/vif_multicollinearity.csv. 20 row(s) shown in full.

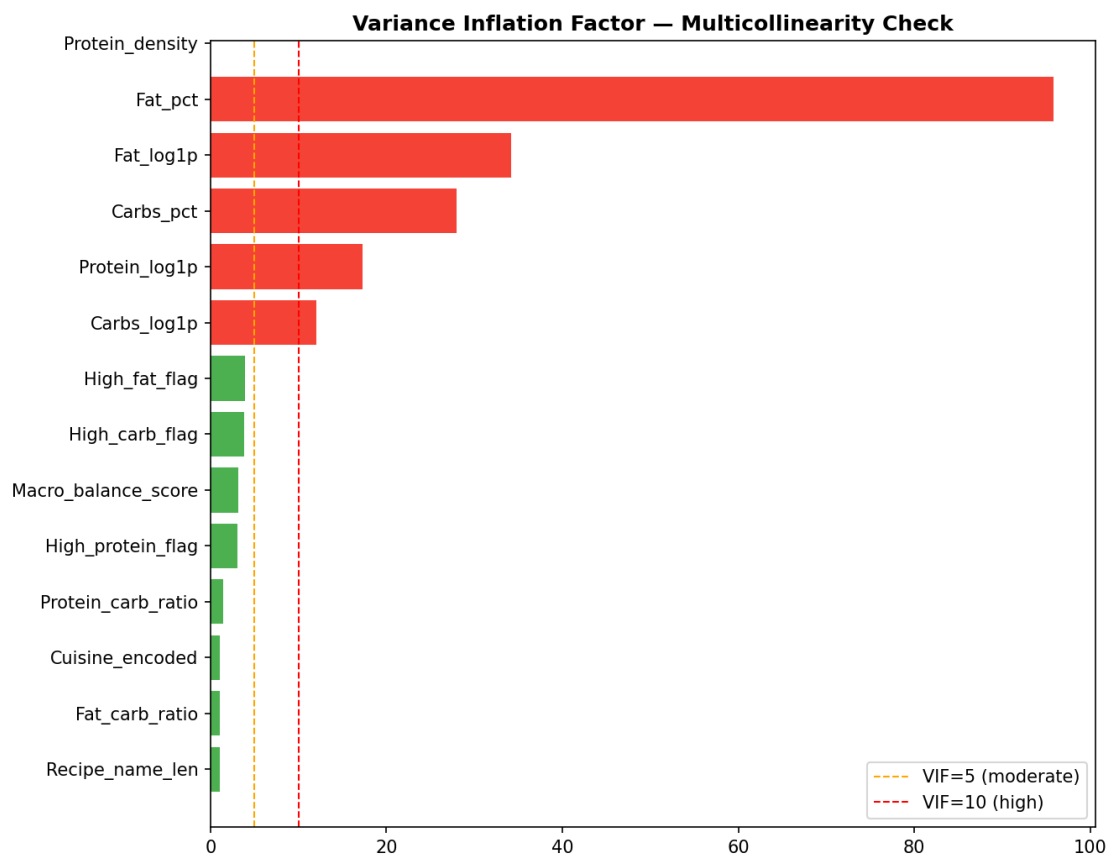


Figure 4.3 — Bar chart of Variance Inflation Factors, with reference thresholds at VIF = 5 and VIF = 10.

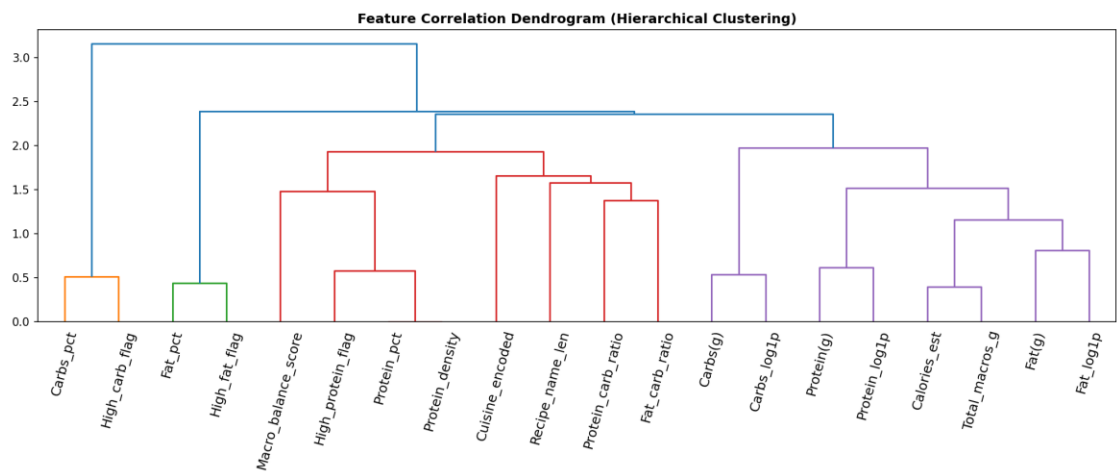


Figure 4.4 — Hierarchical clustering dendrogram of engineered features based on pairwise correlation distance.

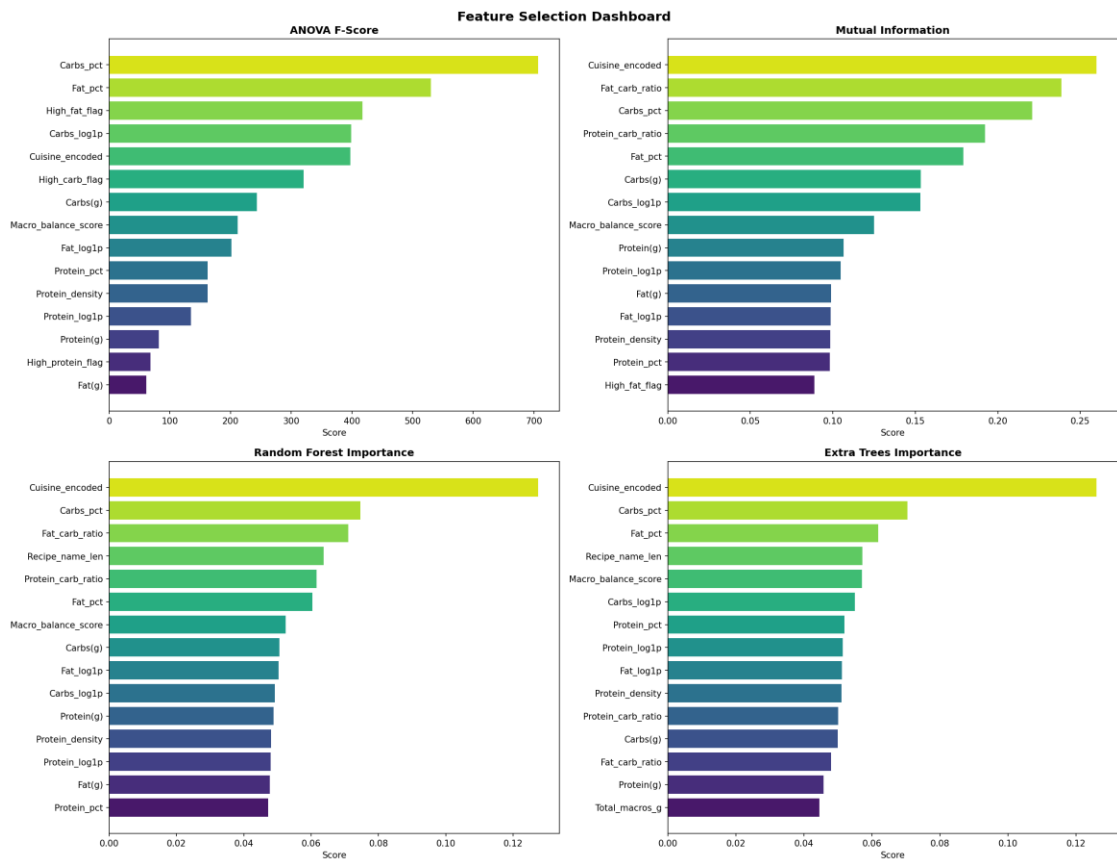


Figure 4.5 — Four-panel comparison of feature-selection methods: ANOVA F-score, mutual information, Random Forest importance, and Extra Trees importance.

Table 4.2 — Consensus Feature Ranking Across Four Selection Methods

	ANOVA_rank	MI_rank	RF_rank	ET_rank	Consensus_rank
Carbs_pct	1.0	3.0	2.0	2.0	2.0
Cuisine_encoded	5.0	1.0	1.0	1.0	2.0

	ANOVA_rank	MI_rank	RF_rank	ET_rank	Consensus_rank
Fat_pct	2.0	5.0	6.0	3.0	4.0
Carbs_log1p	4.0	7.0	10.0	6.0	6.75
Macro_balance_score	8.0	8.0	7.0	5.0	7.0
Carbs(g)	7.0	6.0	8.0	12.0	8.25
Protein_carb_ratio	16.0	4.0	5.0	11.0	9.0
Fat_carb_ratio	20.0	2.0	3.0	13.0	9.5
Fat_log1p	9.0	12.0	9.0	9.0	9.75
Protein_log1p	12.0	10.0	13.0	8.0	10.75
Recipe_name_len	17.0	18.0	4.0	4.0	10.75
Protein_density	10.5	13.0	12.0	10.0	11.375
Protein_pct	10.5	14.0	15.0	7.0	11.625
Protein(g)	13.0	9.0	11.0	14.0	11.75
High_fat_flag	3.0	15.0	18.0	18.0	13.5
Fat(g)	15.0	11.0	14.0	16.0	14.0
High_carb_flag	6.0	16.0	19.0	19.0	15.0
Total_macros_g	18.0	17.0	17.0	15.0	16.75
Calories_est	19.0	19.0	16.0	17.0	17.75
High_protein_flag	14.0	20.0	20.0	20.0	18.5

Source file: features/feature_ranking_consensus.csv. 20 row(s) shown in full.

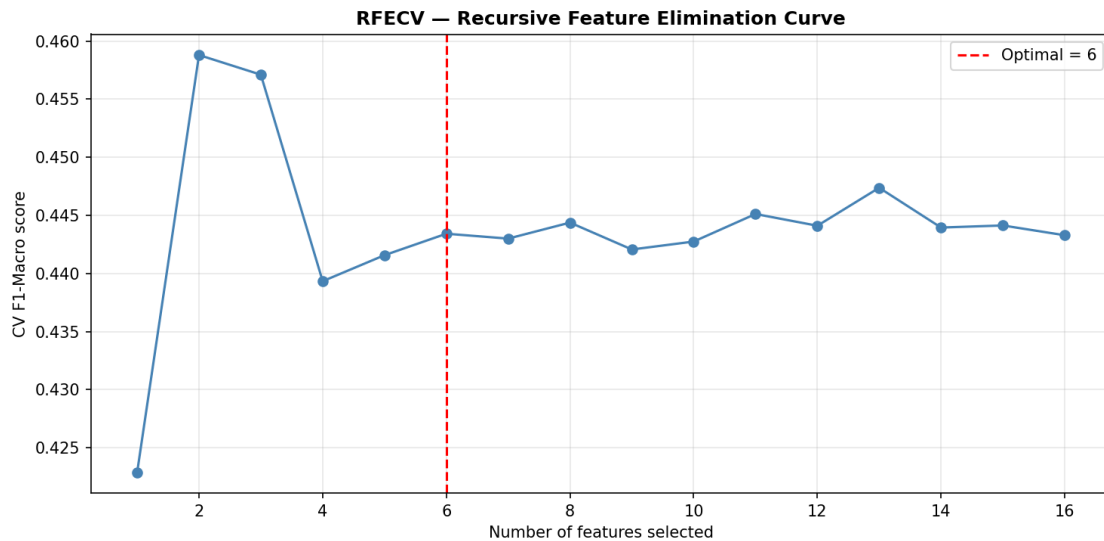


Figure 4.6 — Recursive Feature Elimination with Cross-Validation (RFECV): F1-Macro score as a function of feature-set size.

Note: Following consensus ranking, the top 15 features were retained for model development: Carbs_pct, Cuisine_encoded, Fat_pct, Carbs_log1p, Macro_balance_score, Carbs(g), Protein_carb_ratio, Fat_carb_ratio, Fat_log1p, Protein_log1p, Recipe_name_len, Protein_density, Protein_pct, Protein(g), High_fat_flag.

Table 4.3 — Engineered Master Dataset (Representative Excerpt)

The fully engineered dataset comprises 7,806 records across 36 columns. In accordance with academic and industry reporting conventions, a representative excerpt of the first twelve records is presented below; the complete table is delivered as `df_master_engineered.csv` within the accompanying project archive.

Diet_type	Recipe_name	Cuisine_type	Protein(g)	Carbs(g)	Fat(g)	Calories_est	Protein_pct	Carbs_pct	Fat_pct	Macro_balance_score
paleo	Bone Broth From 'Nom Nom Paleo'	american	5.22	1.29	3.2	54.84	38.074398248758676	9.409190371819673	52.516411377598175	0.3417639038367659
paleo	Paleo Effect Asian-Glazed Pork Sides, A Sweet & Crispy Appetizer	southeast asian	181.55	28.62	146.14	2155.9399999999996	33.68368321936897	5.309980797236793	61.00633598334786	0.16450508763279625
paleo	Paleo Pumpkin Pie	american	30.91	302.59	96.76	2204.84	5.607663141087059	54.89559333101045	39.49674352785713	0.24353839420935885
paleo	Strawberry Guacamole recipes	mexican	9.62	75.78	59.89	880.61	4.369698277325524	34.42159412221707	61.208707600343836	0.14694617348279082
paleo	Asian Cauliflower Fried "Rice" From 'Nom Nom Paleo'	chinese	39.84	54.08	71.55	1019.6299999999999	15.629198827010166	21.21553896999773	63.155262202894036	0.22068546650097987
paleo	Paleo Shrimp-Stuffed Mushrooms recipes	mediterranean	68.62	34.15	42.44	793.04	34.611116715380554	17.224856249347795	48.164027035145565	0.5347265771680909
paleo	Paleo Pumpkin Pie recipes	american	30.03	275.88	97.68	2102.76	5.7124921531674016	52.47959824228514	41.80790960449989	0.264752458918857
paleo	Autoimmune Paleo Pesto	italian	4.25	14.15	72.9	729.7	2.329724544330095	7.7566123064166685	89.91366314911619	-0.4722522753243621
paleo	Baked Banana Chip	french	152.88	1874.52	385.8	11581.800000000001	5.280008288866559	64.74019582447764	29.979795886647153	0.10385167416355101

Diet_type	Recipe_name	Cuisine_type	Protein(g)	Carbs(g)	Fat(g)	Calories_est	Protein_pct	Carbs_pct	Fat_pct	Macro_balance_score
	Encrusted French Toast									
paleo	Vietnamese Pho Pressure Cooker (Noodle Soup)	south east asian	602.91	274.87	400.01	7111.21	33.913215894336695	15.461222492372539	50.625561613276695	0.4723197977513376
paleo	Paleo Collard Burrito	mexican	44.62	28.66	78.46	999.26	17.86121730078472	11.472489642323849	70.66629305679136	0.025338656584126817
paleo	Paleo Chawanmushi (Savory Egg Custard) recipes	american	61.43	13.29	24.6	520.28	47.22841546850306	10.217575151821677	42.55400937948306	0.39535580483374877

Source file: features/df_master_engineered.csv. Showing 12 of 7806 total rows; the complete table is provided in the source CSV file listed above.

7. Data Preprocessing Pipeline

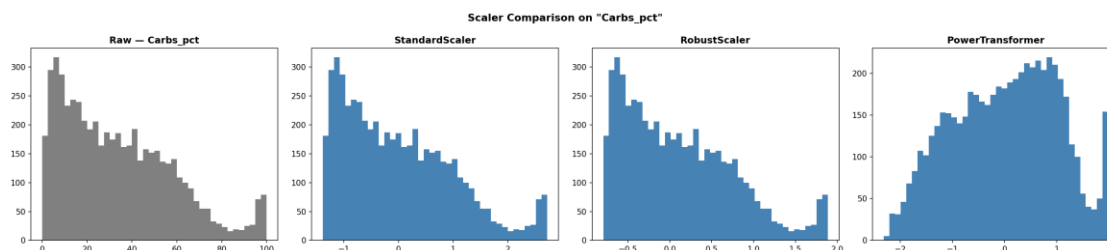


Figure 5.1 — Comparison of raw, standard-scaled, robust-scaled, and power-transformed distributions for a representative engineered feature.

Note: The fitted preprocessing pipeline (`ColumnTransformer`) and the target label encoder were persisted as `preprocessing/preprocessor.joblib` and `preprocessing/label_encoder.joblib` respectively, ensuring that the exact transformation applied during training can be reproduced at inference time.

8. Unsupervised Structure Discovery

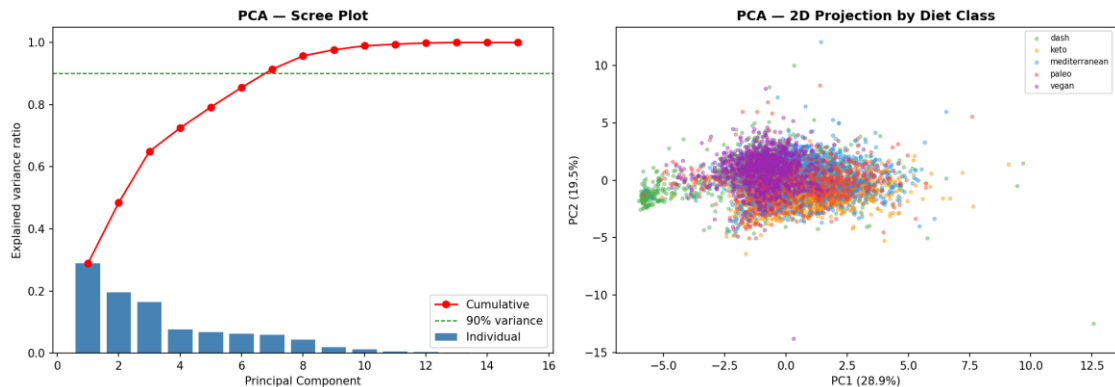


Figure 6.1 — Principal Component Analysis scree plot and two-dimensional projection of the training data, coloured by diet type.



Figure 6.2 — t-distributed Stochastic Neighbour Embedding (t-SNE) of the training feature space.

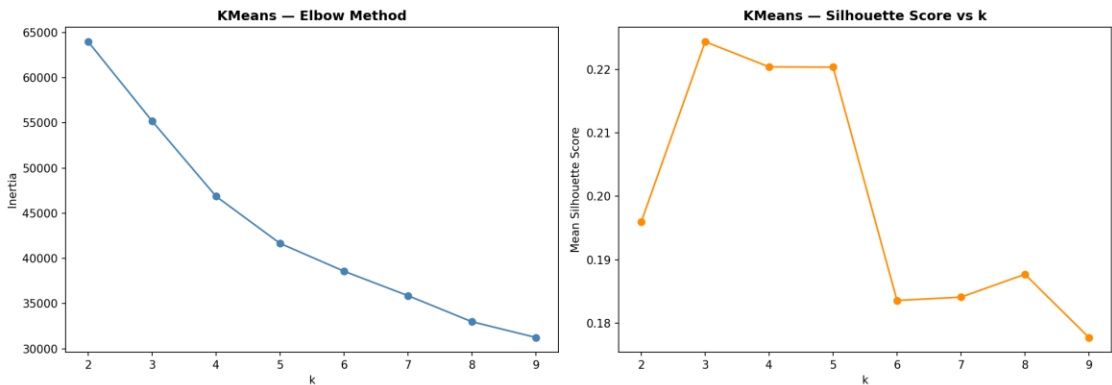


Figure 6.3 — K-Means elbow curve (inertia) and mean silhouette score as a function of the number of clusters, k .

Table 6.1 — K-Means Cluster Assignment ($k = 5$) Versus Ground-Truth Diet Label

True_Diet	0	1	2	3	4
dash	245	496	242	0	238
keto	638	56	9	0	356
mediterranean	269	560	17	0	381
paleo	351	244	33	0	263
vegan	220	784	29	1	31

Source file: unsupervised/kmeans_vs_diet_crosstab.csv. 5 row(s) shown in full.

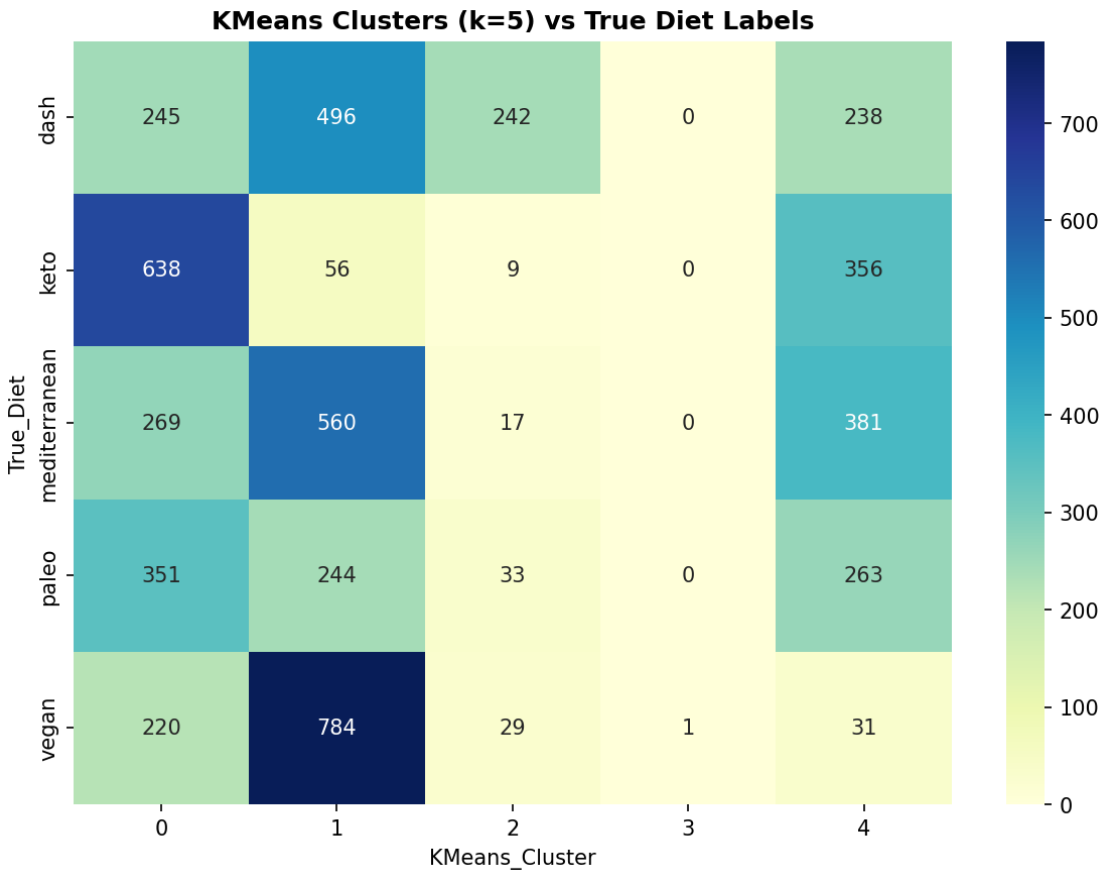


Figure 6.4 — Heatmap of K-Means cluster membership against true diet-type labels.

Note: An interactive three-dimensional PCA projection ([unsupervised/29_pca_3d_interactive.html](#)) was additionally generated and is catalogued in the Artifact Inventory (Section 12).

9. Baseline, Boosting, and Ensemble Modelling

Table 7.1 — Baseline Model Performance (Ten Classical Algorithms)

	Accurac y	F1_Macr o	F1_Weig hted	ROC_AU C	Balanced _Acc	MCC	Kappa	CV_F1_ mean	CV_F1_ st_d	Train _tim e_s
HistGrad ientBoos ting	0.57234 3149807 9385	0.557735 8435634 007	0.566905 9205241 125	0.82635 4772891 292	0.563761 7896117 657	0.465450 4382679 6545	0.464012 4887181 8557	0.536957 0720636 344	0.012416 55173202 96	1.445
Gradient Boosting	0.56658 1306017 9258	0.542264 9139795 287	0.554487 9509973 201	0.83704 9790044 8407	0.554929 2268239 142	0.459207 9328347 282	0.456570 2909565 934	0.539523 7584823 084	0.013969 85111301 078	19.64 4
Random Forest	0.55697 8233034 5711	0.535504 7016621 319	0.544763 6149519 857	0.82313 9965121 4464	0.547127 2156520 052	0.446515 0619762 251	0.444043 4847013 5135	0.524117 4693684 945	0.013957 75018109 2984	5.366
AdaBoos t	0.56081 9462227 9129	0.526169 2083932 812	0.540449 7198438 606	0.80470 1275370 8024	0.547295 5732891 494	0.454394 3331788 8056	0.448667 3996967 375	0.499190 6858894 205	0.006821 71404433 6942	4.845
ExtraTre es	0.53265 0448143 4059	0.511121 9588327 018	0.520126 2688612 005	0.80928 8717818 1263	0.522549 6178936 854	0.415476 1361827 2156	0.413211 6659856 6705	0.512831 4894906 083	0.010833 80563524 9494	1.35
Decision Tree	0.52816 9014084 507	0.510817 4642684 763	0.520645 1185322 181	0.74524 4681236 3062	0.519972 2763233 444	0.411720 5131213 5466	0.409460 5810831 4687	0.492880 6746486 9706	0.005499 58949136 0982	0.114
KNeighb ors	0.51536 4916773 3675	0.494913 6499248 8174	0.503216 4314706 132	0.78156 5561286 2996	0.504237 3521879 545	0.392704 6139476 9834	0.390318 1762870 2903	0.460702 0706007 562	0.011372 25441570 441	0.008
LogisticR egressio n	0.50896 2868117 7977	0.473920 8513102 2643	0.481871 9821237 324	0.78503 6641591 6494	0.501109 1431578 116	0.393157 9259101 153	0.384750 4063391 475	0.460644 9005804 527	0.011019 11644078 045	0.147
RidgeCla ssifier	0.51408 4507042 2535	0.466061 2540125 0686	0.475522 2617085 0537	—	0.504200 8562735 214	0.402686 5807524 684	0.389990 9905228 1205	0.449969 3301774 9496	0.015411 67363641 056	0.006
SGDClas sifier	0.46670 9346991 0371	0.402875 4478751 585	0.416028 8595239 5036	—	0.451713 9121177 7856	0.341036 8397506 721	0.327405 9339815 2207	0.403392 7713007 7233	0.026161 39454451 954	0.111

Source file: models/baseline_results.csv. 10 row(s) shown in full.

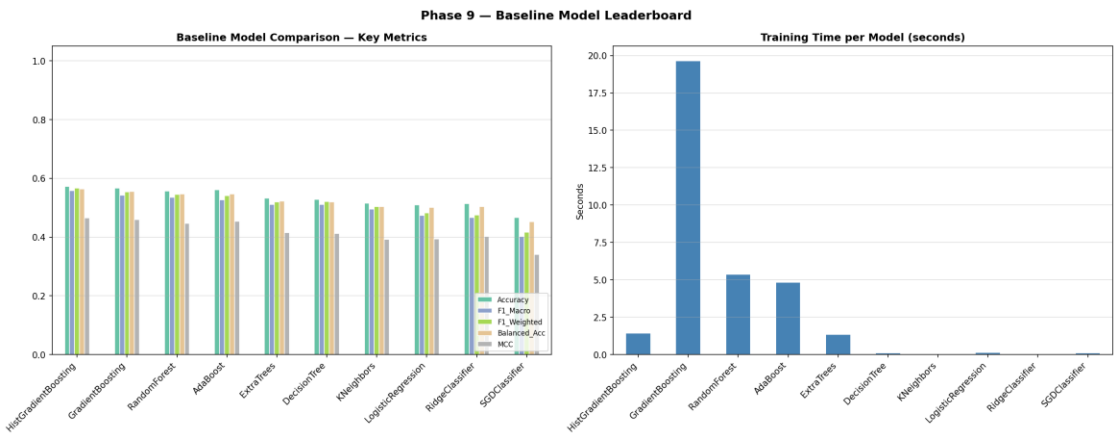


Figure 7.1 — Comparative leaderboard of baseline model metrics and training time.

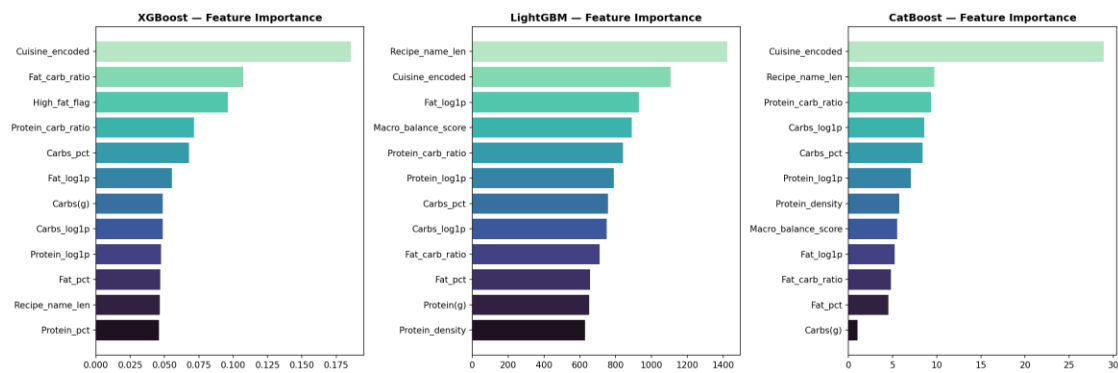


Figure 7.2 — Feature-importance comparison across the XGBoost, LightGBM, and CatBoost gradient-boosting models.

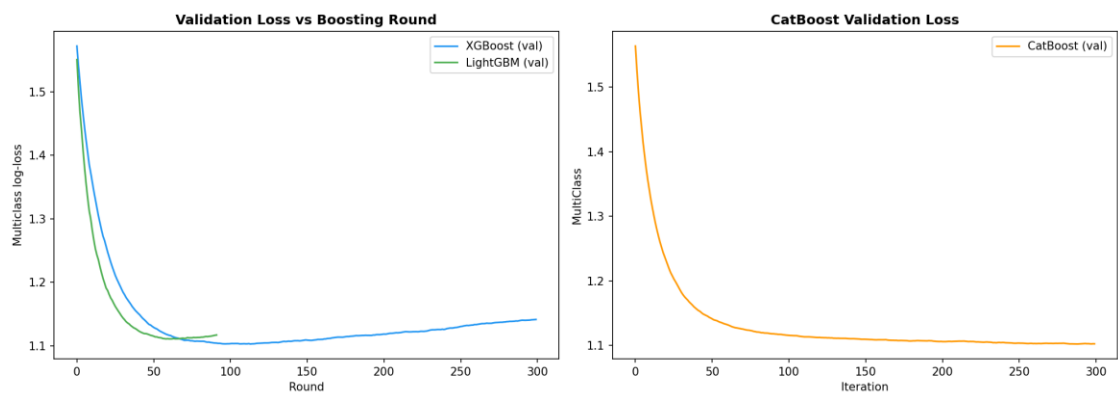


Figure 7.3 — Validation-loss training curves for the gradient-boosting models.

Table 7.2 — Ensemble Versus Individually Tuned Model Comparison

	Accuracy	F1_Macro
XGBoost (tuned)	0.5787451984635084	0.5546213884295419
LightGBM (tuned)	0.5729833546734955	0.5499163827834156
RandomForest (tuned)	0.56145966709347	0.5430446569803762
CatBoost	0.5755441741357235	0.5527140324548137
Voting Ensemble	0.5729833546734955	0.5491772134970729
Stacking Ensemble	0.5729833546734955	0.5541373884775104

Source file: `models/ensemble_comparison.csv`. 6 row(s) shown in full.

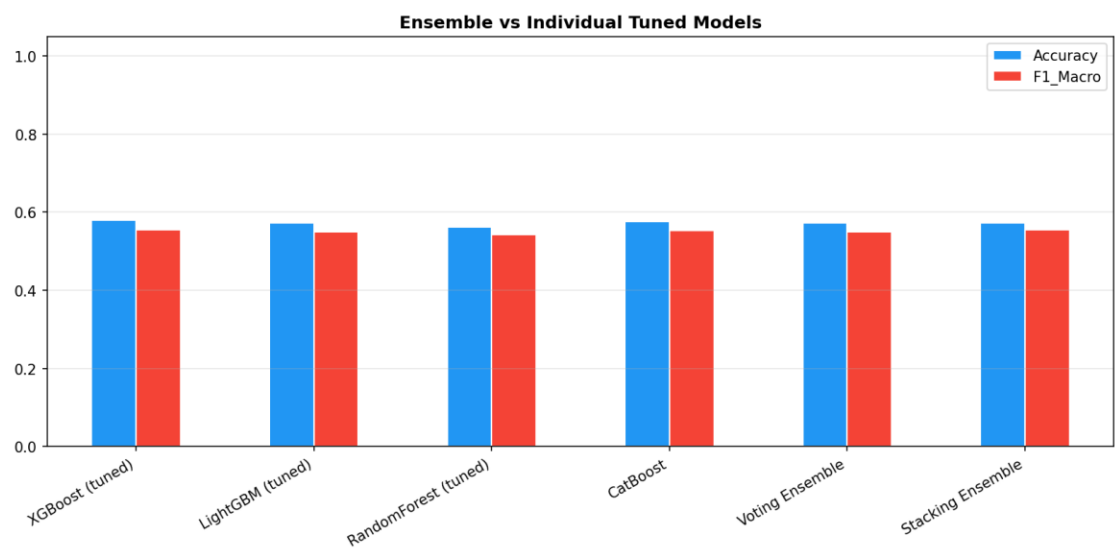


Figure 7.4 — Bar-chart comparison of Voting and Stacking ensembles against their constituent tuned base models.

Table 7.3 — End-to-End Inference Demonstration on Five Held-Out Records

Diet_type	Recipe_name	Cuisine_type	Protein(g)	Carbs(g)	Fat(g)	Predicted_Diet	Confidence
mediterranean	Mediterranean Bean, Potato and Vegetable Salad Platter	mediterranean	42.14	238.2	19.21	vegan	0.6816
dash	Salmon and Asparagus Wraps	nordic	36.32	59.09	12.08	dash	0.6117
dash	Pomegranate Chicken with Preserved Lemon and Almond Rice Pilaf Recipe	mediterranean	280.03	290.25	277.65	dash	0.4028
paleo	Carolina BBQ Meatballs (Low Carb & Glu	american	92.06	28.05	117.08	keto	0.7284
mediterranean	Mediterranean Grilled Whole Snapper with Fennel and a Pernod Butter Sauce	mediterranean	289.65	60.75	116.53	paleo	0.4101

Source file: models/orchestration_demo_predictions.csv. 5 row(s) shown in full.

10. Hyperparameter Optimisation and Regularisation Analysis

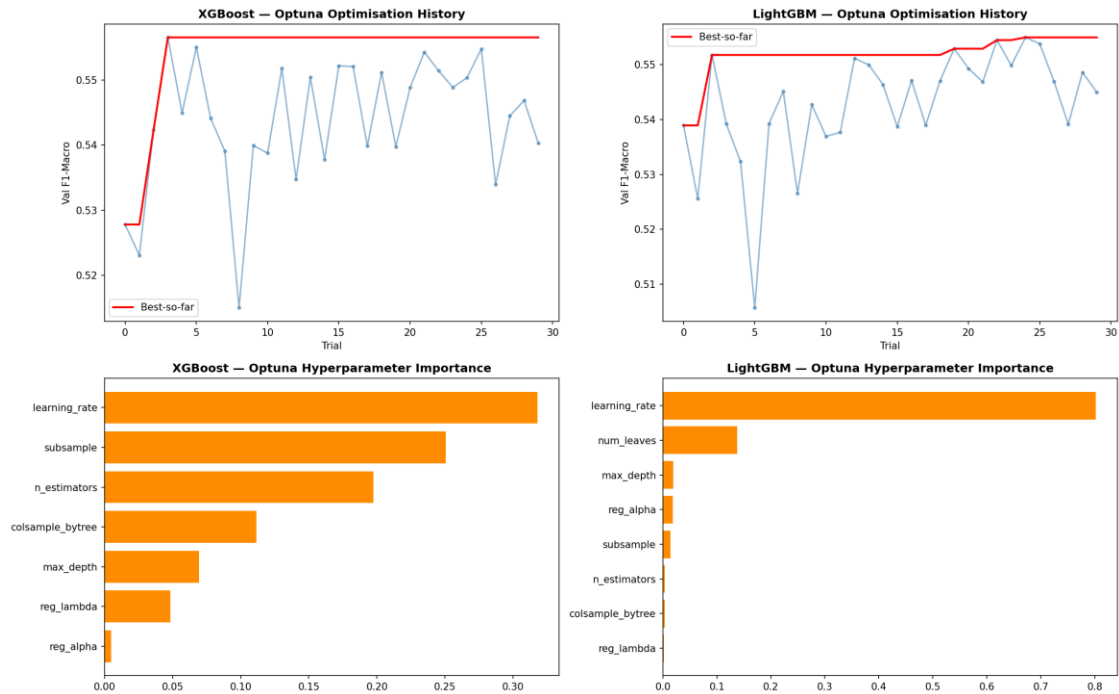


Figure 8.1 — Optuna optimisation-history and hyperparameter-importance diagnostics for the XGBoost and LightGBM tuning studies.

Table 8.1 — Optuna Trial Log — XGBoost (Tree-structured Parzen Estimator, 30 Trials)

number	value	params_max_depth	params_learning_rate	params_n_estimators	params_subsample	params_colsample_bytree
0	0.5278019154020017	10	0.1205712628744377	250	0.8394633936788146	0.6624074561769746
1	0.5230646282203508	7	0.11114989443094977	447	0.608233797718321	0.9879639408647978
2	0.5423629377667163	4	0.028145092716060652	172	0.8099025726528951	0.7727780074568463
3	0.556579099863787	5	0.03476649150592621	155	0.7824279936868144	0.9140703845572055
4	0.5449662099813939	3	0.07896186801026692	337	0.6682096494749166	0.6260206371941118
5	0.5550591215023661	5	0.013940346079873234	424	0.8736932106048627	0.7760609974958406
6	0.5441650726005809	10	0.024112898115291985	113	0.8650089137415928	0.7246844304357644
7	0.5390904229815793	10	0.13962563737015762	174	0.9757995766256756	0.9579309401710595
8	0.5149898038418825	4	0.011662890273931383	135	0.7301321323053057	0.7554709158757928
9	0.5399539915478434	5	0.06333268775321842	243	0.6563696899899051	0.9208787923016158

number	value	params_max_depth	params_learning_rate	params_n_estimators	params_subsample	params_colsample_bytree
10	0.5387713795155811	8	0.27047297227177763	350	0.9729161367647149	0.8683510849293969
11	0.5518045197955244	6	0.010179205671557655	480	0.9029275796221661	0.8462948165495067
12	0.5347415585754481	6	0.025622896109281097	392	0.7643900954643358	0.8355007583653367
13	0.5504479734659306	5	0.043621516342190886	274	0.9087696049520285	0.904488418630764
14	0.5377758012750496	8	0.016298777041229843	423	0.7667698433310003	0.7987586652315627
15	0.5522209445587954	3	0.035727080844059	499	0.9133324745267224	0.6841717712554698
16	0.5520852678169481	5	0.016816570016264992	316	0.8016720883409235	0.9121554425248775
17	0.5398635818271581	6	0.05057894523628074	204	0.7144083225862191	0.7060032876839232
18	0.5511834498335055	4	0.01704786716771474	378	0.8466116826142228	0.6009696054506537
19	0.5397764161571097	7	0.019803593188091745	302	0.9406069544760228	0.8029967056557425
20	0.5488585002911932	7	0.012621616285069909	205	0.867116979833637	0.9582128893978354
21	0.5543062848209412	3	0.03529893812964345	493	0.9092095794740992	0.684887706596041
22	0.5514489221770591	3	0.03712186831324116	453	0.8881926784772247	0.7378578185445576
23	0.5488898758041338	4	0.060603467549399444	423	0.9989918542144398	0.6762744002099903
24	0.55039724228867	5	0.03140898054705142	470	0.8047532884564411	0.795795163509494
25	0.5548133985214183	3	0.0816722574573556	392	0.9449923596722283	0.6404060136851814
26	0.5339221014370195	4	0.18640275615884738	374	0.8331662932409768	0.6422857397373491
27	0.5444998185430745	6	0.07668889769398203	409	0.9534315829598057	0.8900646198797656
28	0.5468796920274969	5	0.09259965073028478	285	0.9438808931823901	0.8313992424127532
29	0.5402998172749026	4	0.14509112062598584	256	0.7663021691191627	0.6493683700479093

Source file: tuning/optuna_xgb_trials.csv. 30 row(s) shown in full.

Table 8.2 — Optuna Trial Log — LightGBM (Tree-structured Parzen Estimator, 30 Trials)

number	value	params_max_depth	params_num_leaves	params_learning_rate	params_n_estimators
0	0.5389413608048841	12	97	0.07661100707771368	250

number	value	params_max_dept h	params_num_leave s	params_learning_rat e	params_n_estimator s
1	0.525561938175301	10	17	0.2708160864249968	341
2	0.5517868160923671	8	63	0.02692655251486473	222
3	0.5392395688987515	10	37	0.05748924681991978	282
4	0.5323484846599333	12	124	0.1563510870813346	126
5	0.5056809984531382	7	18	0.22038218939289875	148
6	0.5392678213930936	4	124	0.13962563737015762	319
7	0.5451531848302344	4	20	0.03023795012558475	135
8	0.5265613723940994	8	30	0.1530883741573138	212
9	0.5427143084097178	11	94	0.11935262708510537	102
10	0.5369495093413983	6	56	0.010629965824084656	478
11	0.5376712349802591	3	60	0.0249178303129701	195
12	0.5511675076962123	5	60	0.02628774929234595	183
13	0.5499888798273379	6	68	0.021652690184947147	385
14	0.5463796319080061	8	86	0.013467002568954148	199
15	0.5387083467890024	5	77	0.03897803960375351	252
16	0.5471111989599399	9	47	0.01723909390602947	179
17	0.5389757706207827	7	44	0.05490605884701748	409
18	0.5470738575131878	3	71	0.03756997754990156	239
19	0.5529258216555971	5	105	0.017398788842188345	306
20	0.5493092881096858	8	108	0.016603487037250227	306
21	0.5468765204751874	5	107	0.022498588228949346	367
22	0.5544976105147295	5	80	0.010011249462227543	278
23	0.5498685672462928	6	82	0.010180985879263062	275
24	0.5549916382752259	4	112	0.01513239336127886	446

number	value	params_max_depth	params_num_leaves	params_learning_rate	params_n_estimators
25	0.5538156820518148	4	111	0.014144241505949328	500
26	0.5469741638078992	4	117	0.013011450354504414	488
27	0.5391482416681351	3	115	0.013334264629651258	454
28	0.548619501869577	4	93	0.010123857592947703	436
29	0.5449921316662735	3	96	0.019065921028478035	432

Source file: tuning/optuna_lgb_trials.csv. 30 row(s) shown in full.

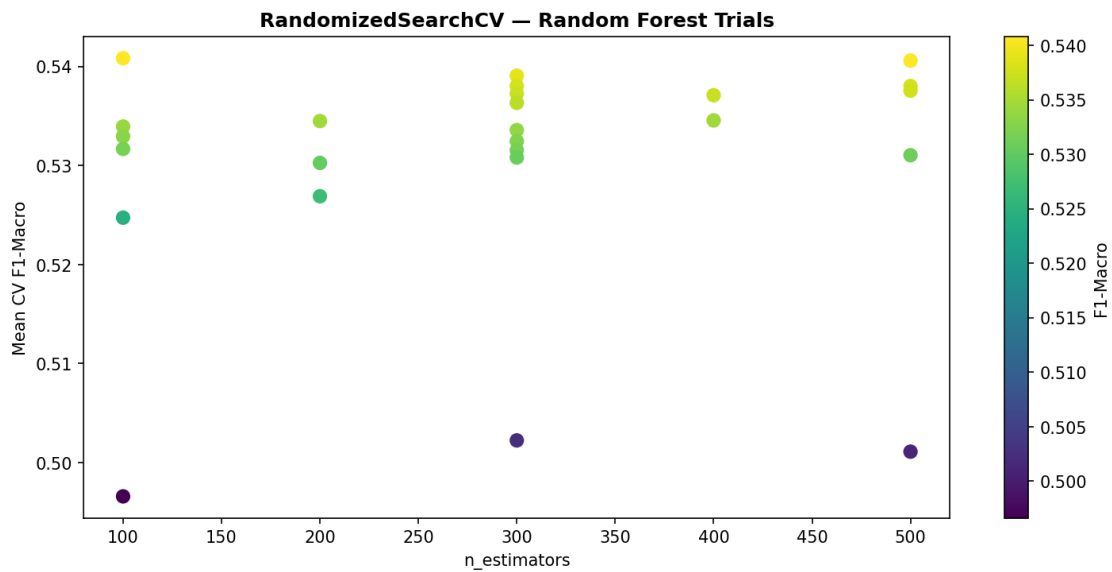


Figure 8.2 — RandomizedSearchCV trial scatter for the Random Forest classifier.

Table 8.3 — Top 15 RandomizedSearchCV Configurations — Random Forest (Ranked by CV F1-Macro)

param_n_estimators	param_max_depth	param_min_samples_split	param_min_samples_leaf	param_max_features	mean_test_score	std_test_score	rank_test_score
100	10	10	1	—	0.5408236439210811	0.010096346464946998	1
500	—	10	1	—	0.5405866110937824	0.010766185242170325	2
300	10	2	4	—	0.5390577994829927	0.0066181950812801964	3
300	15	2	4	sqrt	0.5380197122396806	0.01069859142140217	4
500	15	2	4	sqrt	0.5380133274352507	0.013534265228642338	5
500	15	10	4	sqrt	0.5375490272613763	0.012785194814976279	6

param_n_estimators	param_max_depth	param_min_samples_split	param_min_samples_leaf	param_max_features	mean_test_score	std_test_score	rank_test_score
300	20	10	2	log2	0.537263715391245	0.01348136285470838	7
400	—	10	4	—	0.5370887587113141	0.010672230646833734	8
300	—	2	4	—	0.5363221283881804	0.00825243566677311	9
400	—	5	1	—	0.534545698009377	0.01043483884871942	10
200	15	5	4	—	0.534481258784359	0.006957991005868812	11
100	20	2	4	log2	0.5339274539428394	0.01067210406691514	12
300	15	2	2	—	0.5335648964327834	0.00885552937394585	13
100	10	10	4	log2	0.5329297911061684	0.012718272900244586	14
300	10	10	1	log2	0.5324503607565076	0.012839464744857256	15

Source file: tuning/rf_randomsearch_results.csv. Showing 15 of 25 total rows; the complete table is provided in the source CSV file listed above.

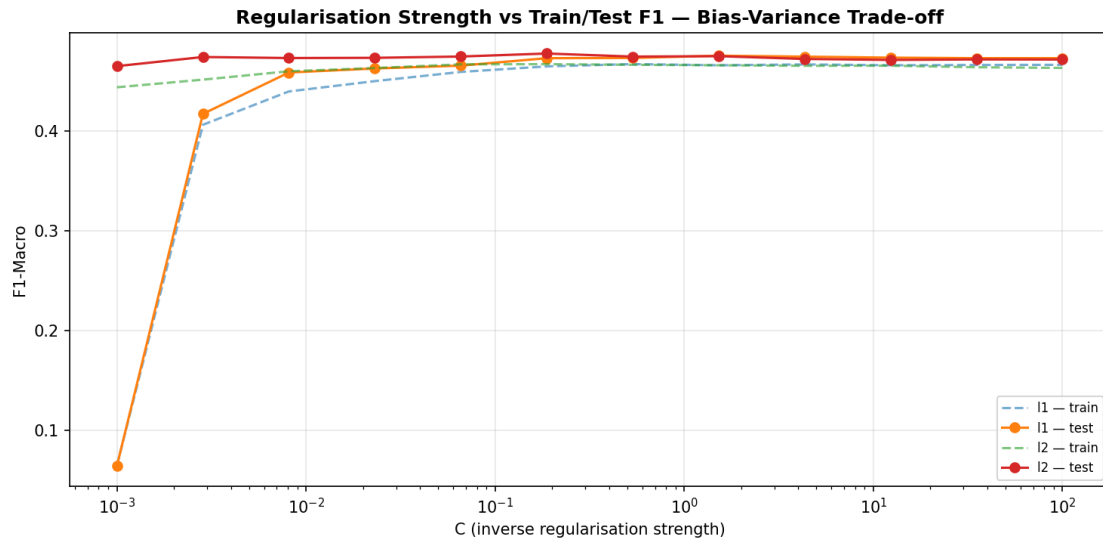


Figure 8.3 — Regularisation strength (C) versus train/test F1-Macro for L1- and L2-penalised logistic regression, illustrating the bias–variance trade-off.

Table 8.4 — Regularisation Strength Sweep — Train/Test F1-Macro

C	penalty	train_f1	test_f1
0.001	l1	0.06494940202391905	0.0648068669527897
0.001	l2	0.443655961113664	0.4648298388658748
0.002848035868435802	l1	0.4061174195973646	0.4170359426149819
0.002848035868435802	l2	0.4513234057895844	0.4739877761901994
0.008111308307896872	l1	0.4395030191959366	0.45845958271332377

C	penalty	train_f1	test_f1
0.008111308307896872	l2	0.45969287659539226	0.4729586456682407
0.02310129700083159	l1	0.4498285339703081	0.4625019318199475
0.02310129700083159	l2	0.4629468676553209	0.4731668679052515
0.0657933224657568	l1	0.4590396535439646	0.4653460813690031
0.0657933224657568	l2	0.4666525476180857	0.4744604764122375
0.1873817422860383	l1	0.4645764163709162	0.47281094135823754
0.1873817422860383	l2	0.4667952677961395	0.47747002756092377
0.5336699231206307	l1	0.46688461343477455	0.47300156738210736
0.5336699231206307	l2	0.46618711083409947	0.4744051575647255
1.5199110829529332	l1	0.4657177319132873	0.4753054884876288
1.5199110829529332	l2	0.46568548380654873	0.47480193157334954
4.328761281083057	l1	0.4665543733821151	0.47434984244101697
4.328761281083057	l2	0.4651839075383535	0.4720312762532958
12.32846739442066	l1	0.46572860617564726	0.4733399598376076
12.32846739442066	l2	0.4653014451275774	0.47106763054297807
35.11191734215127	l1	0.46599717894676707	0.47283275449657547
35.11191734215127	l2	0.46372032137392577	0.4715489358631884
100.0	l1	0.46613894517813426	0.47283275449657547
100.0	l2	0.4630379445538386	0.47143953092133817

Source file: tuning/regularisation_study.csv. 24 row(s) shown in full.

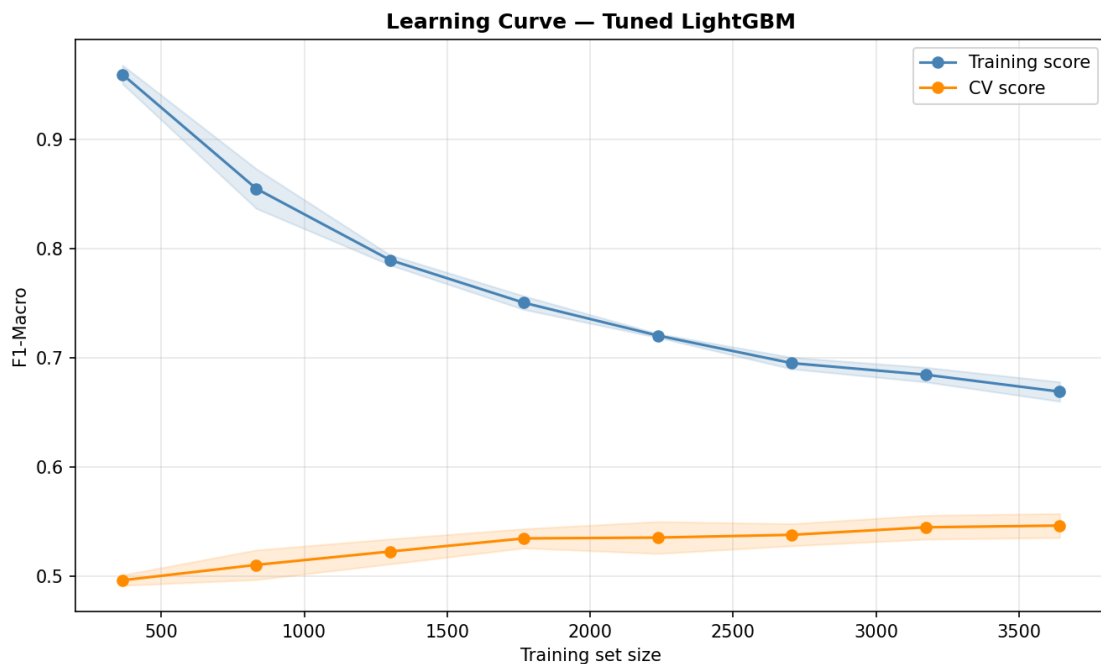


Figure 8.4 — Learning curve (training-set size versus F1-Macro) for the tuned LightGBM model.

11. Comprehensive Model Evaluation

Table 9.1 — Master Leaderboard: All Sixteen Models Ranked by F1-Macro

	Accuracy	F1_Macro	F1_Weighted	ROC_AUC	Balanced_Acc	MCC	Kappa	LogLoss
HistGradientBoosting	0.5723431498079385	0.5577358435634007	0.5669059205241125	0.826354772891292	0.5637617896117657	0.46545043826796545	0.46401248871818557	1.1705897659966058
XGBoost_tuned	0.5787451984635084	0.5546213884295419	0.5661572099923394	0.8415015016280875	0.5679943390212661	0.4757322217556881	0.4719727837335339	1.0720987857812494
StackingEnsemble	0.5729833546734955	0.5541373884775104	0.5652911090424031	0.8390064959935675	0.5624052492782987	0.4663459954397766	0.4646641684252498	1.0858067857810327
CatBoost	0.5755441741357235	0.5527140324548137	0.5631664083245215	0.8447312758469958	0.5656866285551688	0.4716698740849058	0.4678915298338173	1.0632442749365625
LightGBM_tuned	0.5729833546734955	0.5499163827834156	0.5608976988905818	0.8395841110701378	0.5625815906058106	0.46804059678390464	0.4646935994689345	1.0747457185873674
VotingEnsemble	0.5729833546734955	0.5491772134970729	0.5603243252301561	0.8441717845536835	0.5628663735969346	0.46880660418806835	0.464956959461472	1.0631757355470206
RandomForest_tuned	0.56145966709347	0.5430446569803762	0.5529924155367859	0.8344234202880012	0.5537452561236895	0.45444080585624036	0.4513092538519683	1.1126645386857903
GradientBoosting	0.5665813060179258	0.5422649139795287	0.5544879509973201	0.8370497900448407	0.5549292268239142	0.4592079328347282	0.4565702909565934	1.0832983736066988
RandomForest	0.5569782330345711	0.5355047016621319	0.5447636149519857	0.8231399651214464	0.5471272156520052	0.4465150619762251	0.44404348470135135	1.1741613865647769
AdaBoost	0.5608194622279129	0.5261692083932812	0.5404497198438606	0.8047012753708024	0.5472955732891494	0.45439433317888056	0.4486673996967375	1.5933385100821646
ExtraTrees	0.5326504481434059	0.5111219588327018	0.5201262688612005	0.8092887178181263	0.5225496178936854	0.41547613618272156	0.41321166598566705	1.7591282778645547
DecisionTree	0.528169014084507	0.5108174642684763	0.5206451185322181	0.7452446812363062	0.5199722763233444	0.41172051312135466	0.40946058108314687	6.120143296683817
KNeighbors	0.5153649167733675	0.49491364992488174	0.5032164314706132	0.7815655612862996	0.5042373521879545	0.39270461394769834	0.39031817628702903	4.960403943341301
LogisticRegression	0.5089628681177977	0.47392085131022643	0.4818719821237324	0.7850366415916494	0.5011091431578116	0.3931579259101153	0.3847504063391475	1.2390330985026237
RidgeClassifier	0.5140845070422535	0.46606125401250686	0.47552226170850537	—	0.5042008562735214	0.4026865807524684	0.38999099052281205	—
SGDClassifier	0.4667093469910371	0.4028754478751585	0.41602885952395036	—	0.45171391211777856	0.3410368397506721	0.32740593398152207	—

Source file: evaluation/master_leaderboard.csv. 16 row(s) shown in full.

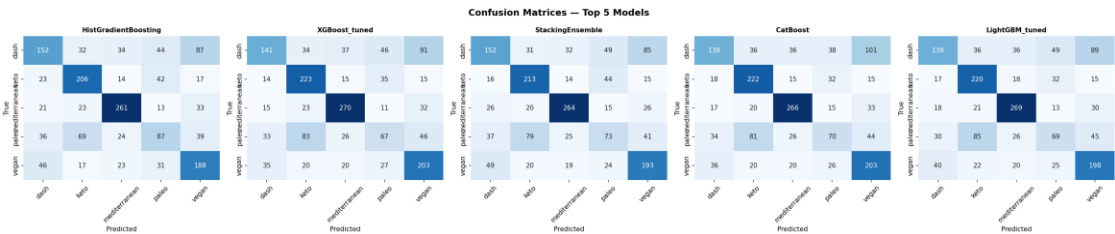


Figure 9.1 — Confusion matrices for the five top-ranked models on the held-out test set.

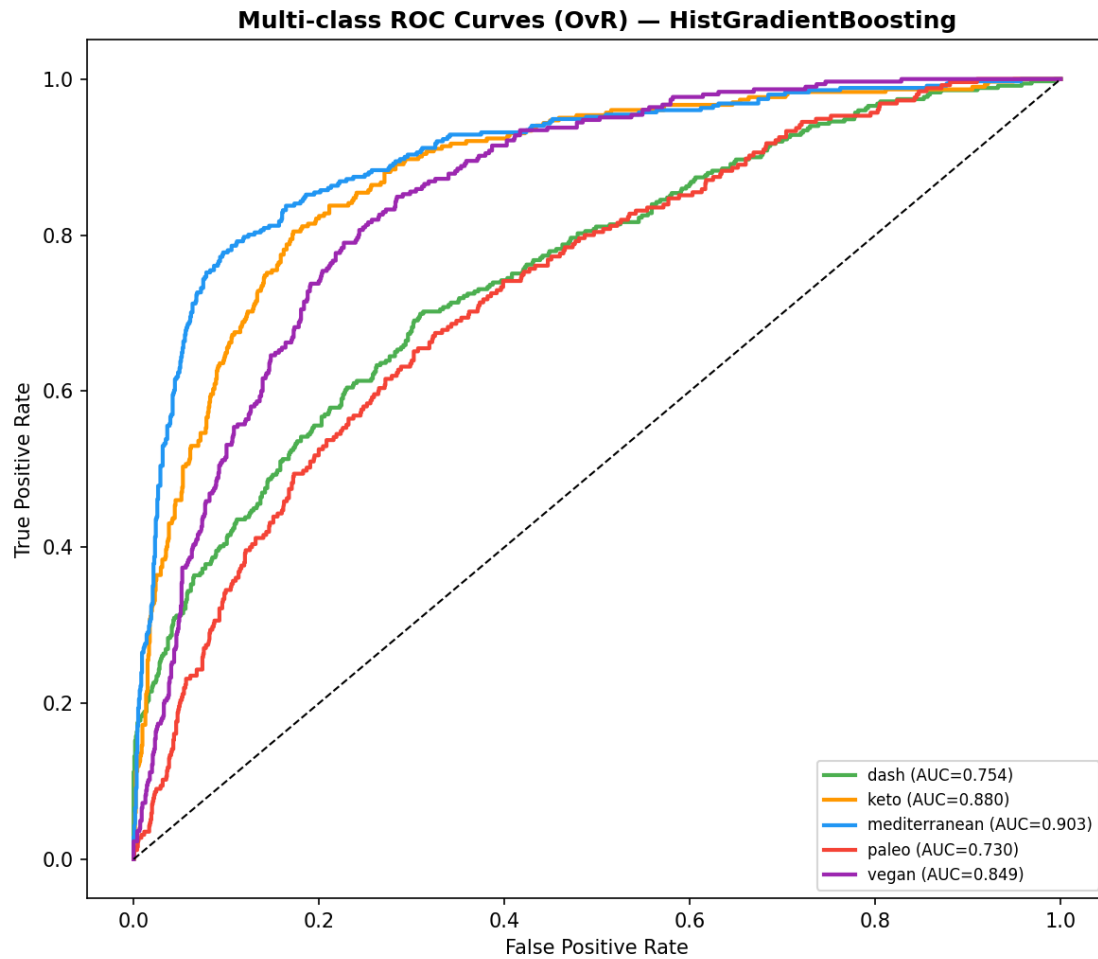


Figure 9.2 — One-versus-rest multi-class ROC curves for the best-performing model.

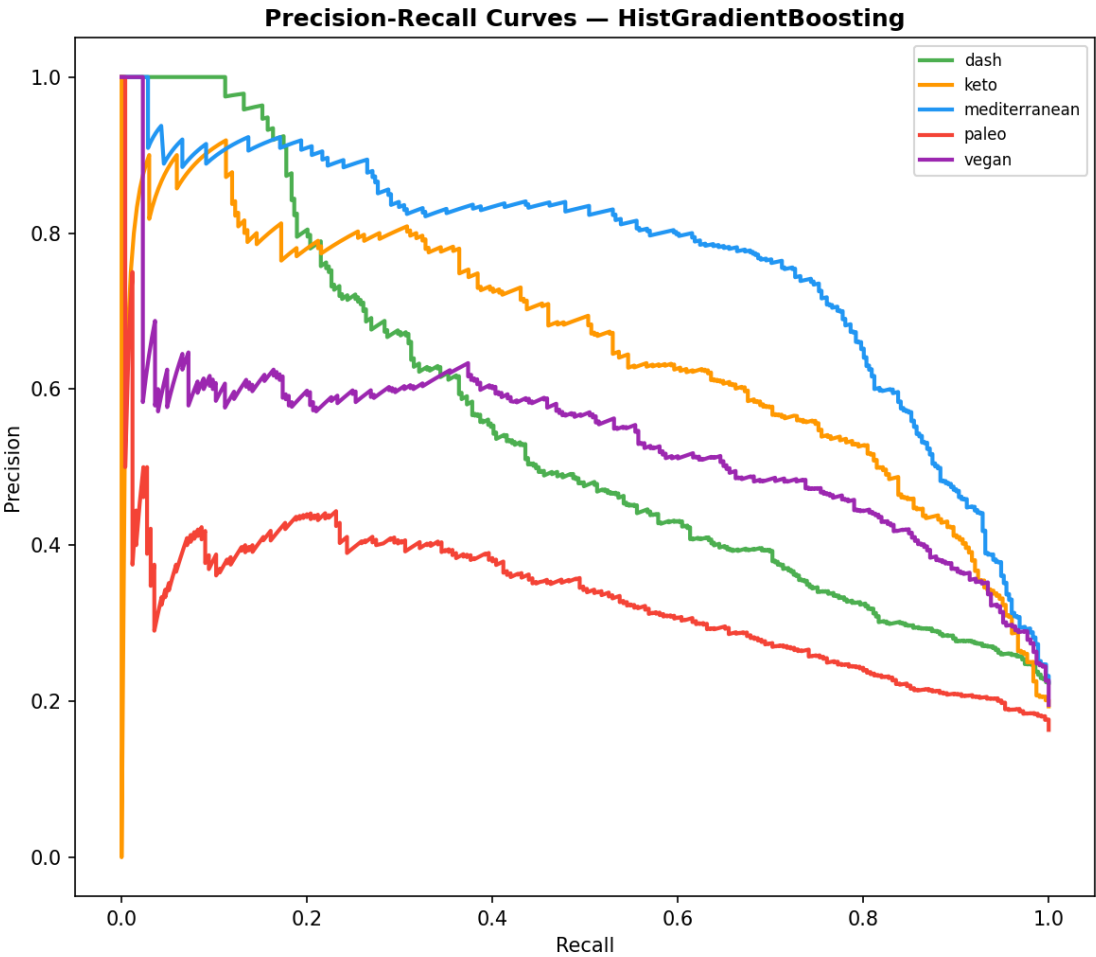


Figure 9.3 — Precision–recall curves per diet class for the best-performing model.

Table 9.2 — Per-Class Precision, Recall, and F1-Score (Best Model)

	precision	recall	f1-score
dash	0.5467625899280576	0.4355300859598854	0.48484848484848486
keto	0.5936599423631124	0.6821192052980133	0.6348228043143297
mediterranean	0.7331460674157303	0.7435897435897436	0.7383309759547383
paleo	0.4009216589861751	0.3411764705882353	0.3686440677966102
vegan	0.5164835164835165	0.6163934426229508	0.5620328849028401

Source file: evaluation/per_class_metrics.csv. 5 row(s) shown in full.

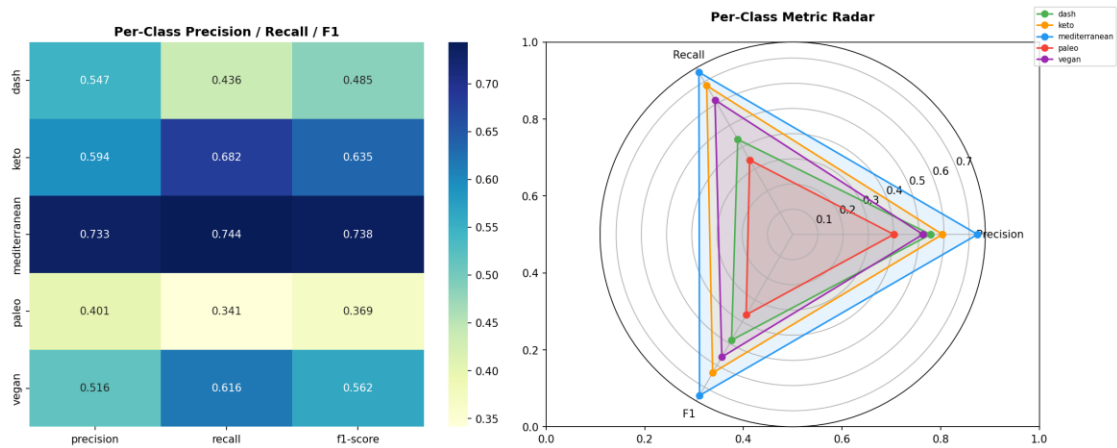


Figure 9.4 — Heatmap and radar-chart visualisation of per-class precision, recall, and F1-score.

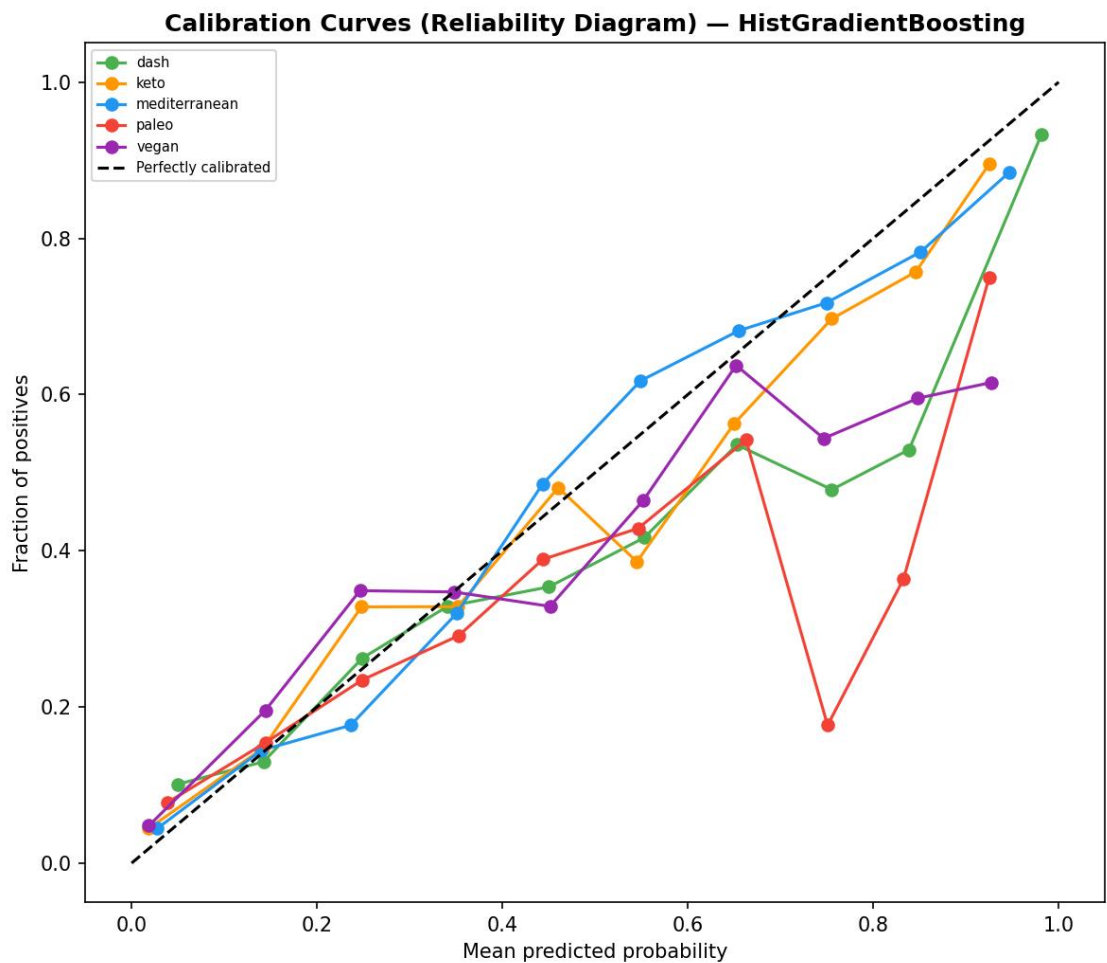


Figure 9.5 — Calibration (reliability) curves for each diet class, best-performing model.

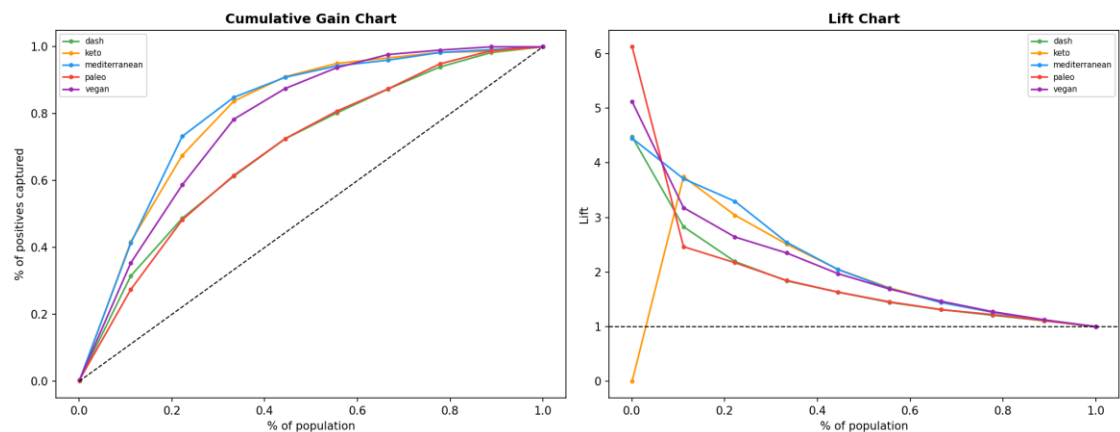


Figure 9.6 — Cumulative gain and lift charts, computed one-versus-rest for each diet class.

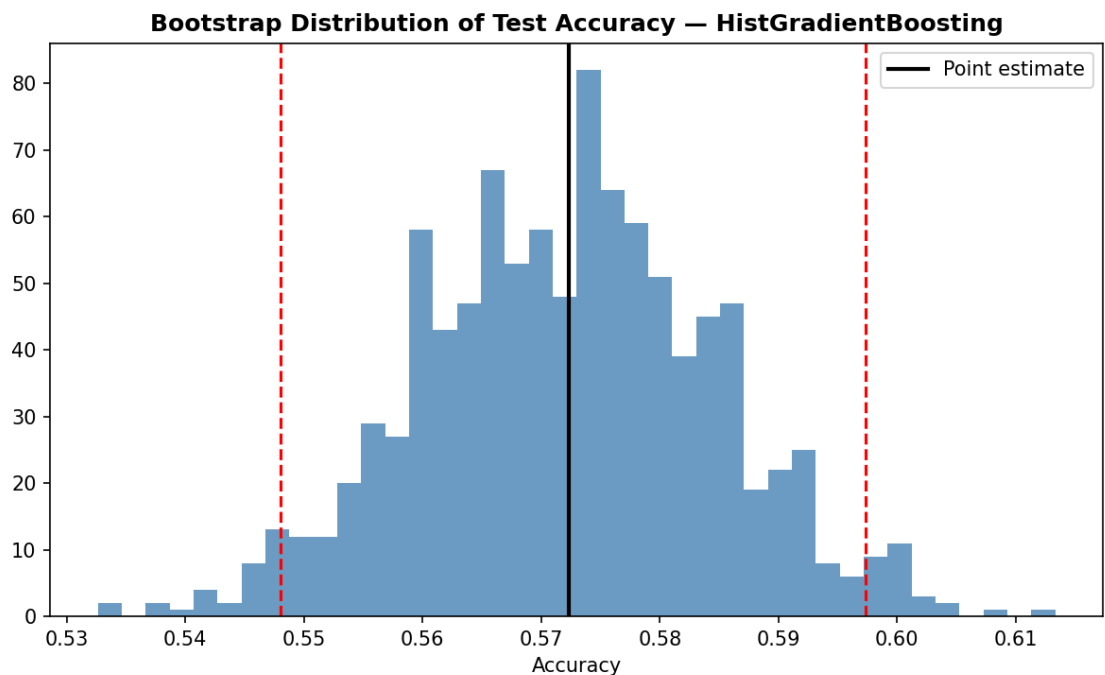


Figure 9.7 — Bootstrap resampling distribution (1,000 iterations) of test-set accuracy for the best-performing model, with the ninety-five percent confidence interval marked.

Table 9.3 — Misclassified Test Records (Representative Excerpt)

668 of the 1,562 held-out test records (42.8%) were misclassified by the best-performing model. A representative excerpt of fifteen misclassified records is presented below; the complete listing is delivered as `evaluation/misclassified_examples.csv`.

Recipe_name	Cuisine_type	Diet_type	Protein(g)	Carbs(g)	Fat(g)	Predicted
Healthy Coconut Meringue Parfait recipes	french	paleo	63.04	161.99	13.34	vegan
5-Ingredient Roasted Tomato	eastern europe	paleo	7.68	30.35	36.87	keto

Recipe_name	Cuisine_type	Diet_type	Protein(g)	Carbs(g)	Fat(g)	Predicted
Soup (Low Carb, Gluten-free) recipes						
BBQ Chicken Sweet Potato Pizza	italian	paleo	112.99	285.71	98.15	mediterranean
Spicy Seasoned Keto Zucchini Fries Recipe	south american	keto	31.24	114.41	19.94	dash
Keto Diet Sloppy Joe Recipe	french	keto	79.29	120.71	91.06	dash
Paleo Shepherd's Pie	american	paleo	124.01	230.07	181.24	dash
Chicken Skillet with Bacon Brussels Sprouts (Paleo, Whole30 + Keto)	italian	keto	284.3	60.25	99.57	paleo
Keto Egg Salad	american	keto	89.52	26.35	153.53	dash
Oven-Baked Parmesan Green Bean Fries	italian	paleo	67.07	37.59	43.58	dash
Dutch Baby Takes a Mediterranean Holiday	american	mediterranean	46.03	103.49	99.18	vegan
Easy Vegan Bacon Bits	american	vegan	59.11	49.68	0.38	dash
Old-Time Beef Stew	american	dash	200.99	61.12	72.78	paleo
Parmesan Herb Chicken (Grain Free and Low Carb)	italian	paleo	166.91	3.84	72.13	keto
Chunky Guacamole (Paleo, Whole30 + Keto)	mexican	keto	21.59	103.56	118.8	vegan
Meat Lite: Eggplant Lamb Lavash Wraps Recipe	american	mediterranean	119.73	253.72	95.35	vegan

Source file: evaluation/misclassified_examples.csv. Showing 15 of 668 total rows; the complete table is provided in the source CSV file listed above.

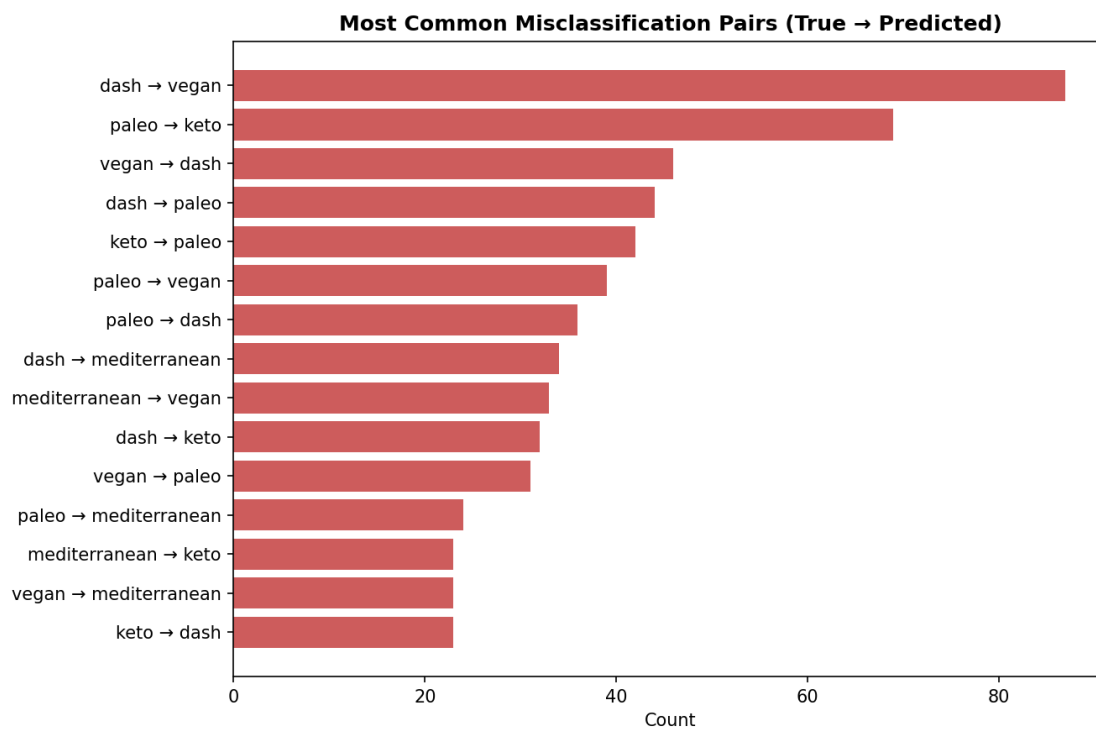


Figure 9.8 — The fifteen most frequent misclassification pairs (true class → predicted class).

Note: An interactive HTML leaderboard chart (evaluation/leaderboard_interactive.html) was also generated and is catalogued in the Artifact Inventory (Section 12).

12. Explainable Artificial Intelligence — SHAP

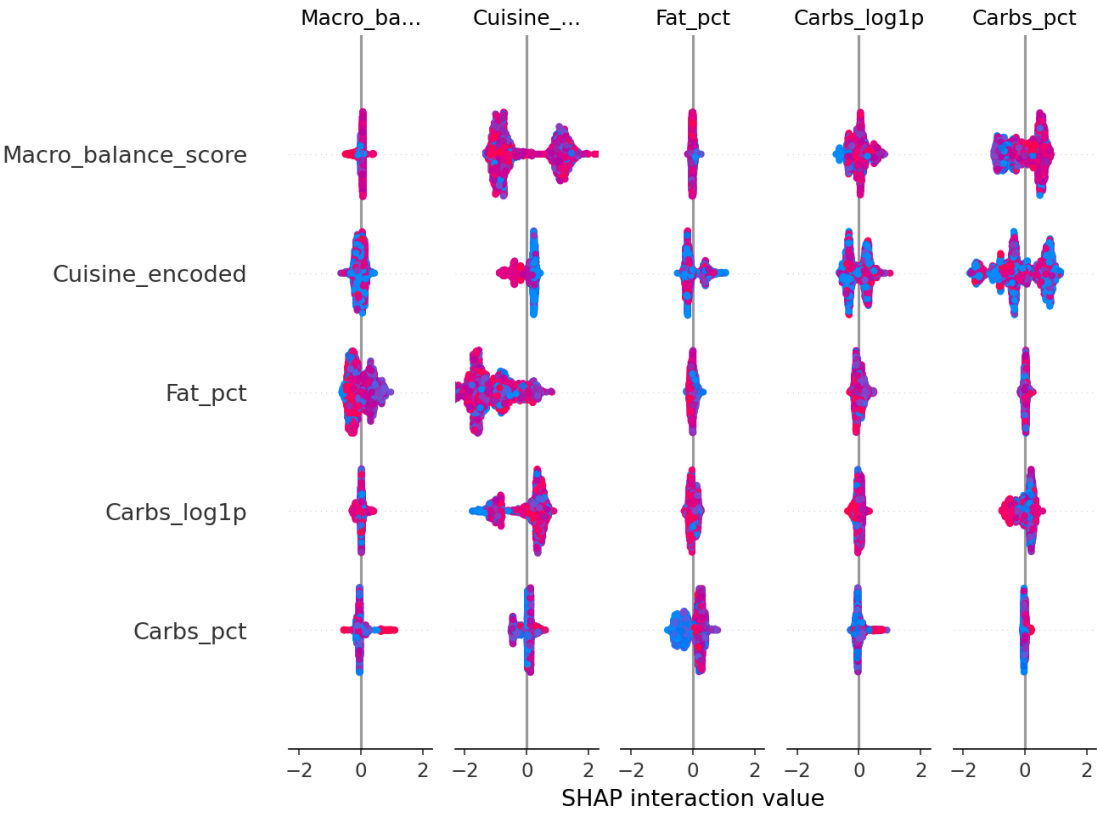


Figure 10.1 — SHAP global summary (beeswarm) plot, tuned LightGBM model, all diet classes.

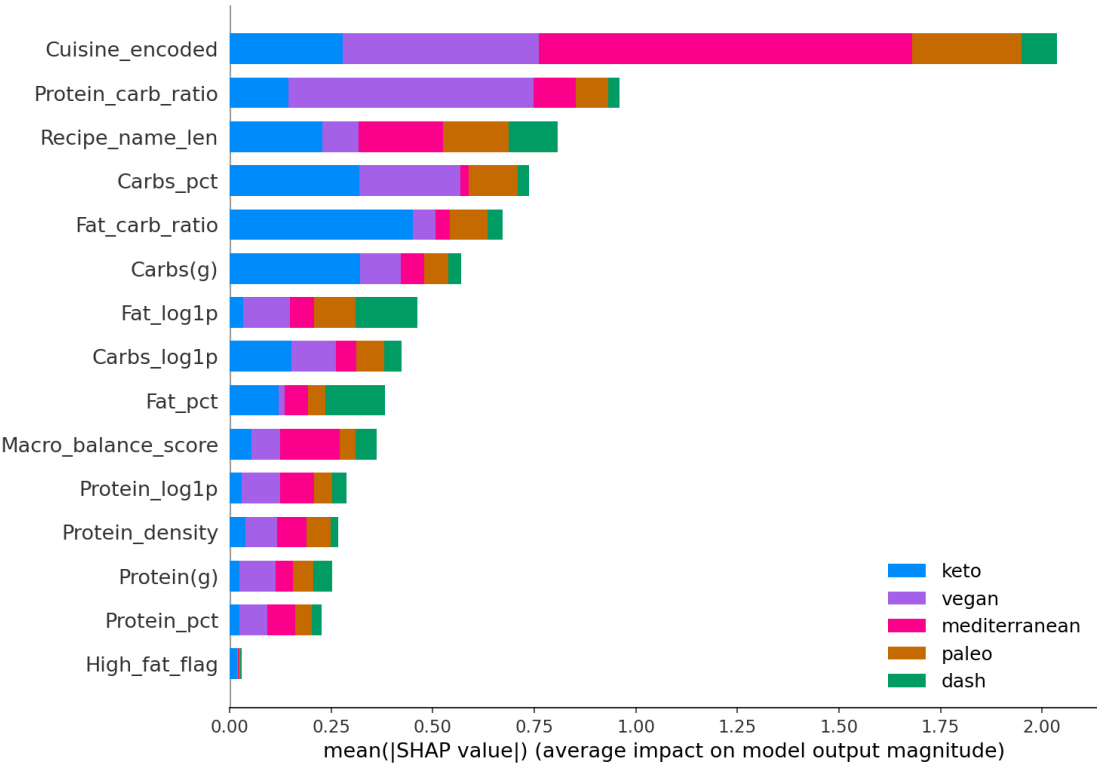


Figure 10.2 — SHAP mean absolute value bar-chart importance, aggregated across all diet classes.

Table 10.1 — SHAP Feature Importance Ranking

Feature	Mean SHAP value
Cuisine_encoded	0.4072036064945861
Protein_carb_ratio	0.19198055981357243
Recipe_name_len	0.1613657869634728
Carbs_pct	0.14748386109333156
Fat_carb_ratio	0.1345106090201608
Carbs(g)	0.11415379777731911
Fat_log1p	0.09233848677269942
Carbs_log1p	0.08480143083343031
Fat_pct	0.07652049450135841
Macro_balance_score	0.07250306405893221
Protein_log1p	0.057673892042982074
Protein_density	0.05344397009741375
Protein(g)	0.050333909138976896
Protein_pct	0.04513604746960108
High_fat_flag	0.0058922129319287685

Source file: xai_shap/shap_feature_importance.csv. 15 row(s) shown in full.

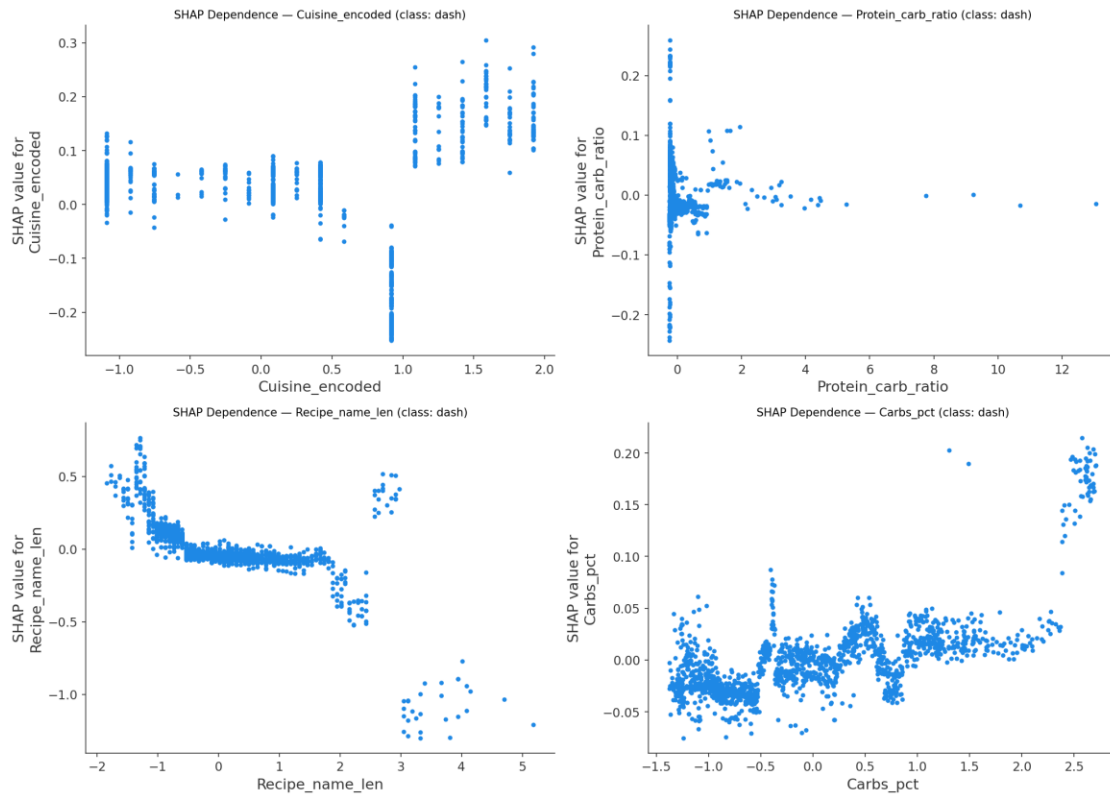


Figure 10.3 — SHAP dependence plots for the four highest-ranked features.

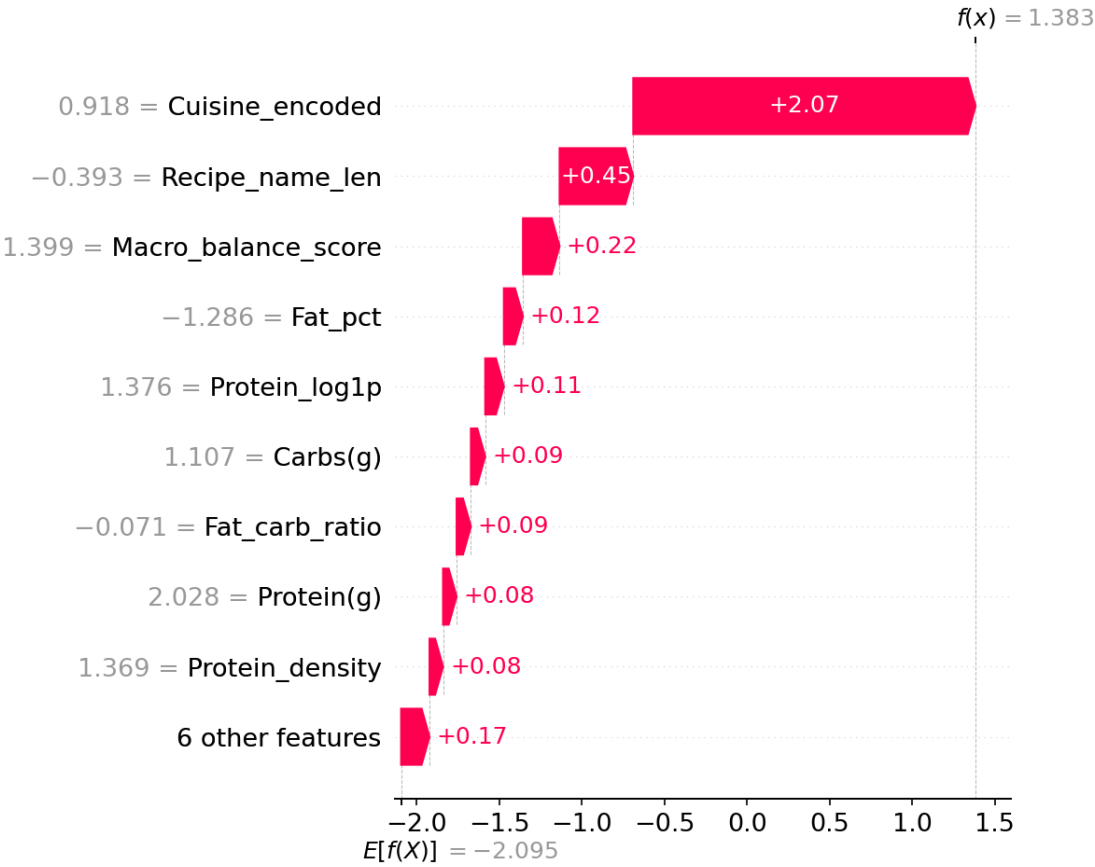


Figure 10.4a — SHAP local waterfall explanation, test sample 0.

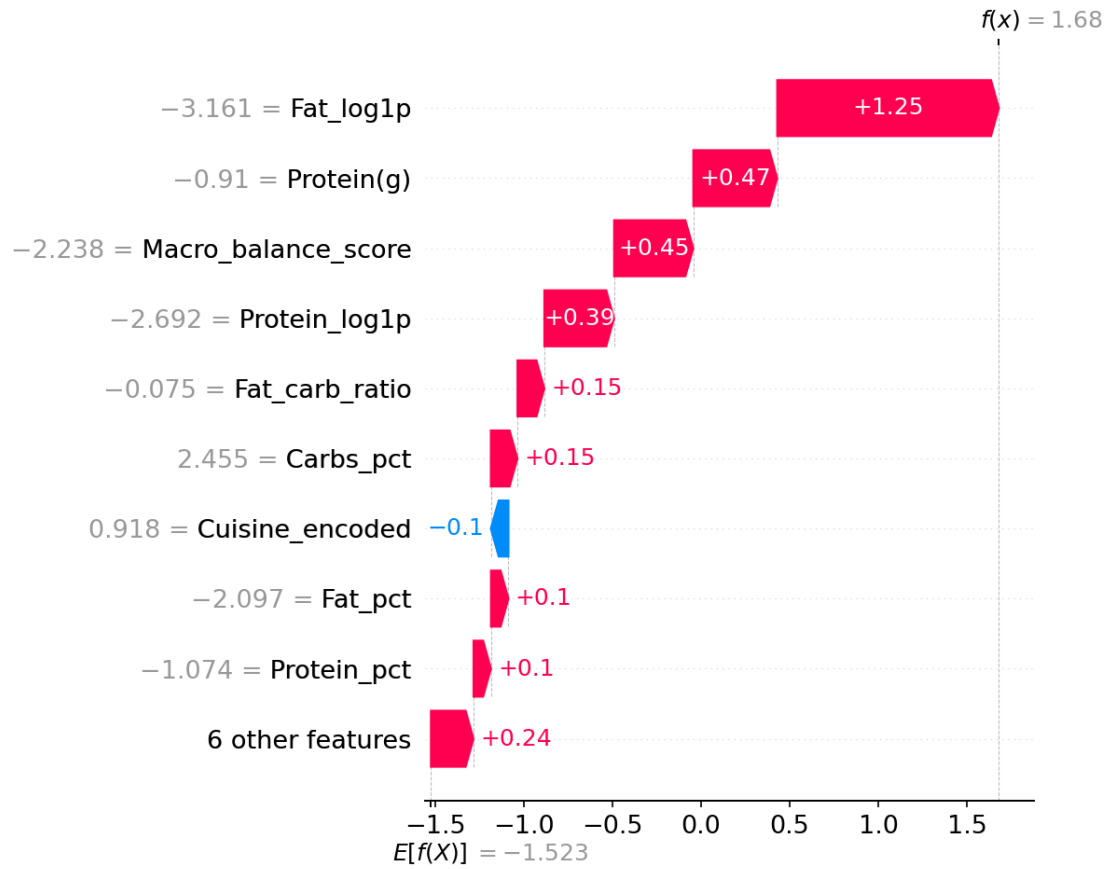


Figure 10.4b — SHAP local waterfall explanation, test sample 1.

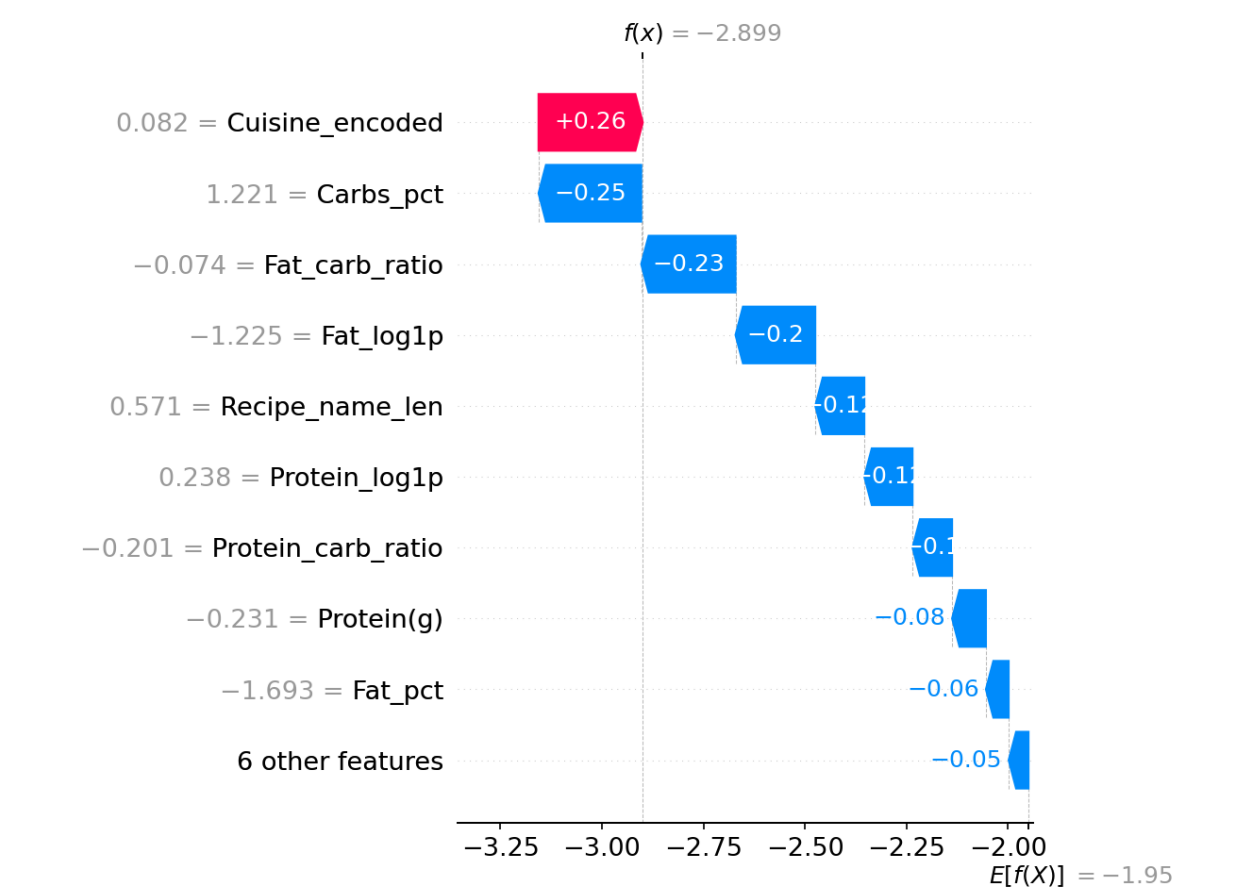


Figure 10.4c — SHAP local waterfall explanation, test sample 2.

Table 10.2 — Permutation Feature Importance (Model-Agnostic Cross-Check)

Feature	Mean importance
Cuisine_encoded	0.1715164857101841
Recipe_name_len	0.014251760025709766
Protein_carb_ratio	0.012125871066688387
Fat_pct	0.007181095960615891
Macro_balance_score	0.005160719251188772
Fat_carb_ratio	0.004499228964058899
Protein(g)	0.004121348220027439
Fat_log1p	0.002189139669692941
Carbs(g)	0.001888619302347879
Carbs_log1p	0.0017434924134849335
Carbs_pct	0.0005647777130171875
Protein_log1p	9.807183806098685e-05
High_fat_flag	-0.0005103192982517306
Protein_pct	-0.00245981959618126
Protein_density	-0.00377732883345215

Source file: xai_shap/permutation_importance.csv. 15 row(s) shown in full.

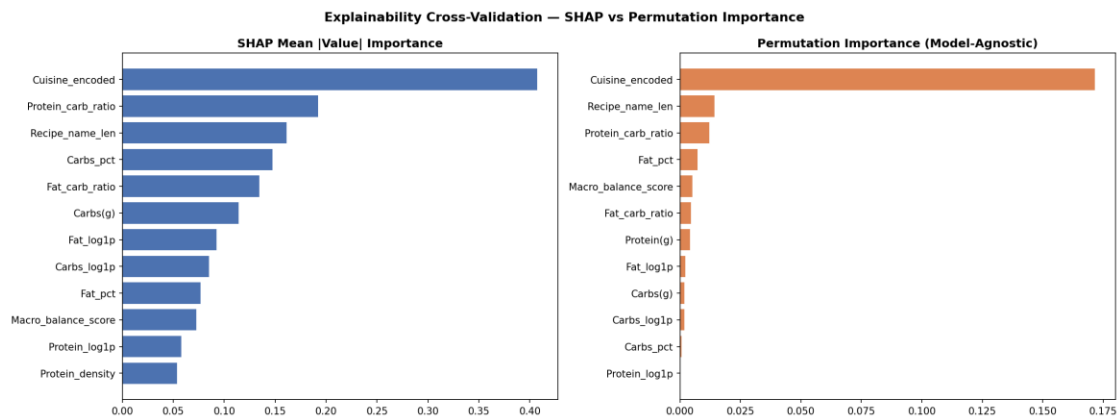


Figure 10.5 — Side-by-side comparison of SHAP importance and permutation importance.

13. Explainable Artificial Intelligence — LIME

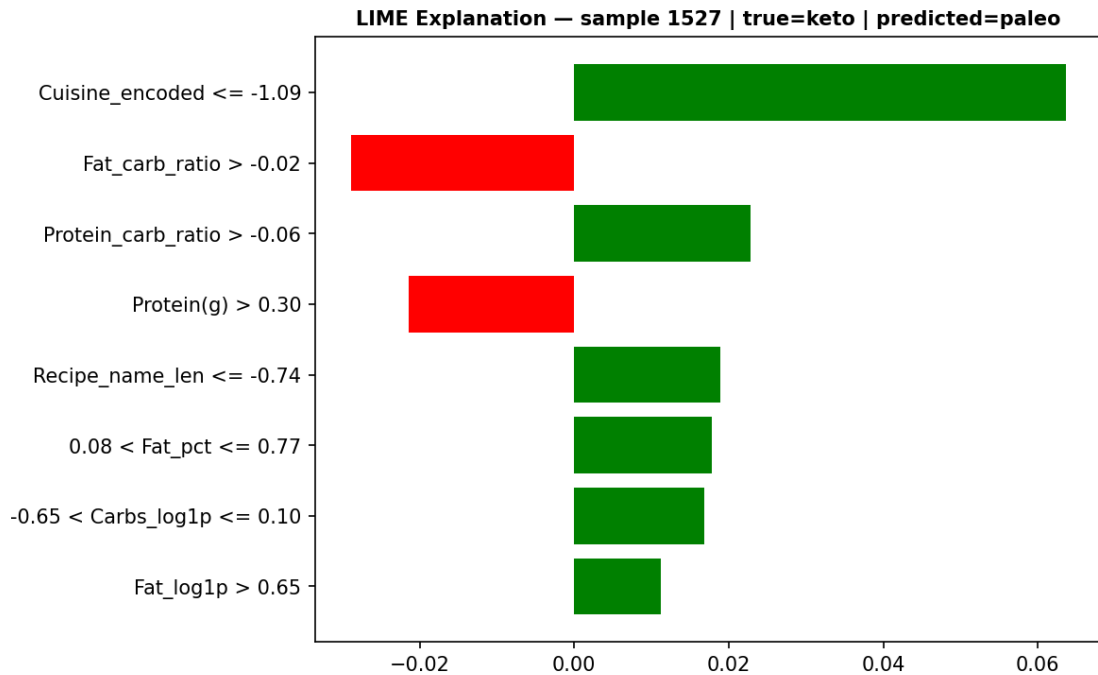


Figure 11.1 — LIME local explanation, test sample 0.

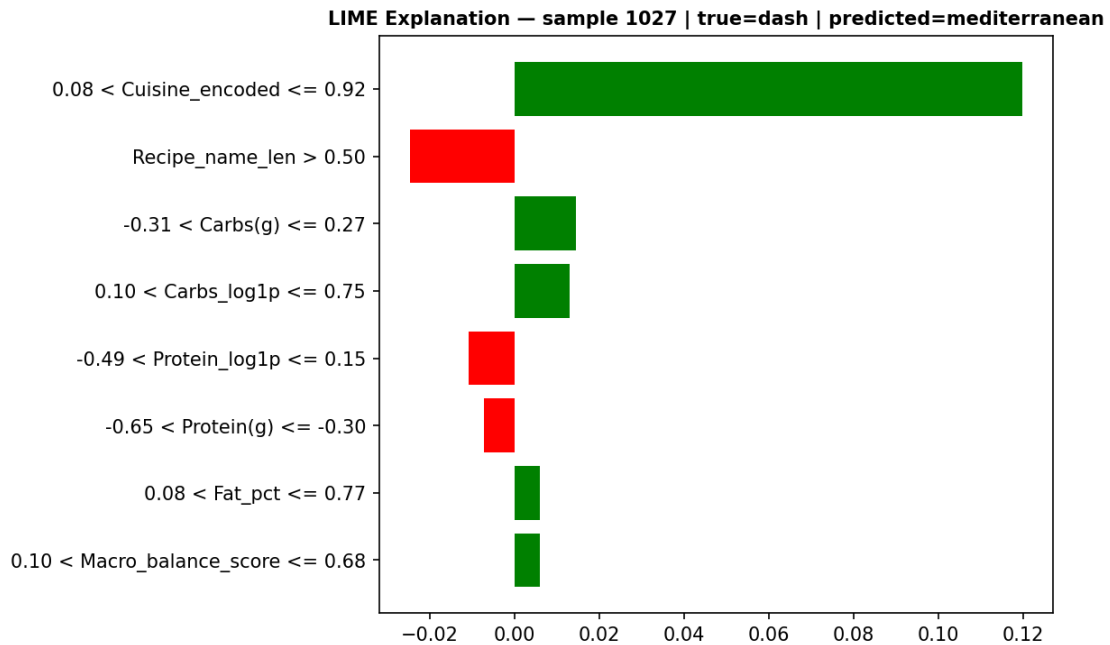


Figure 11.2 — LIME local explanation, test sample 1.

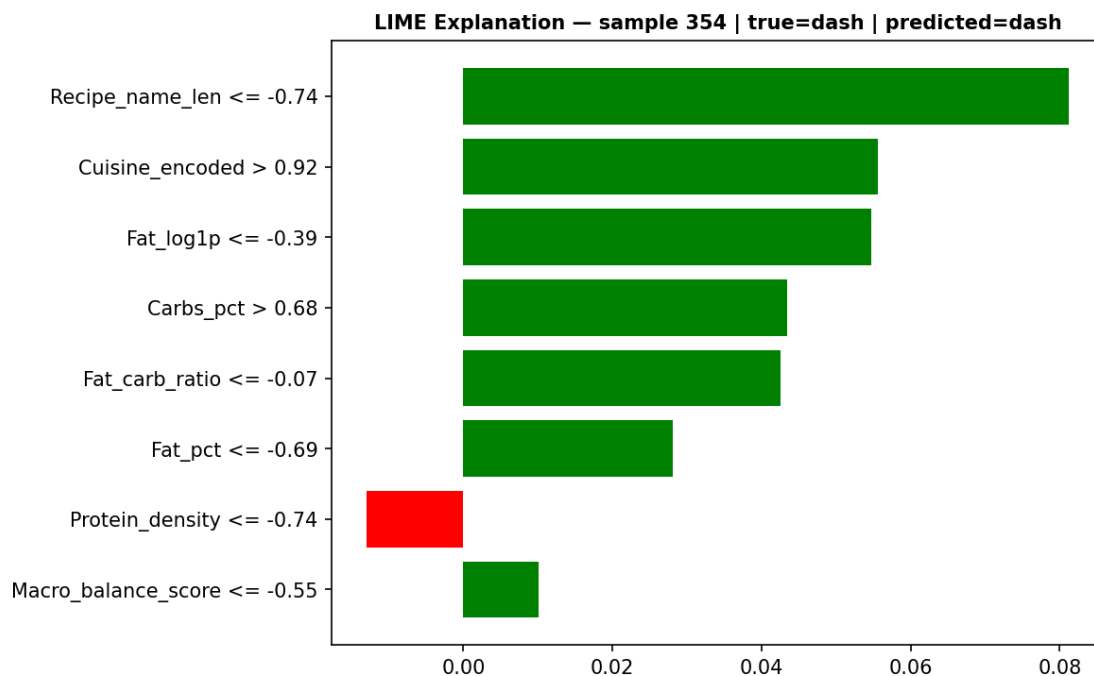


Figure 11.3 — LIME local explanation, test sample 2.

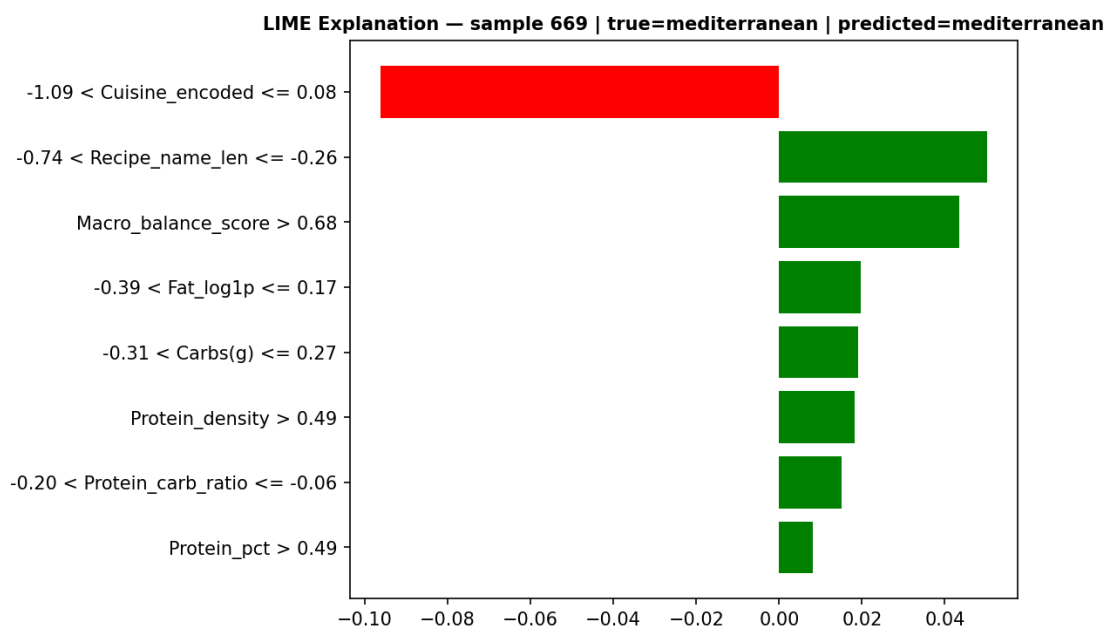


Figure 11.4 — LIME local explanation, test sample 3.

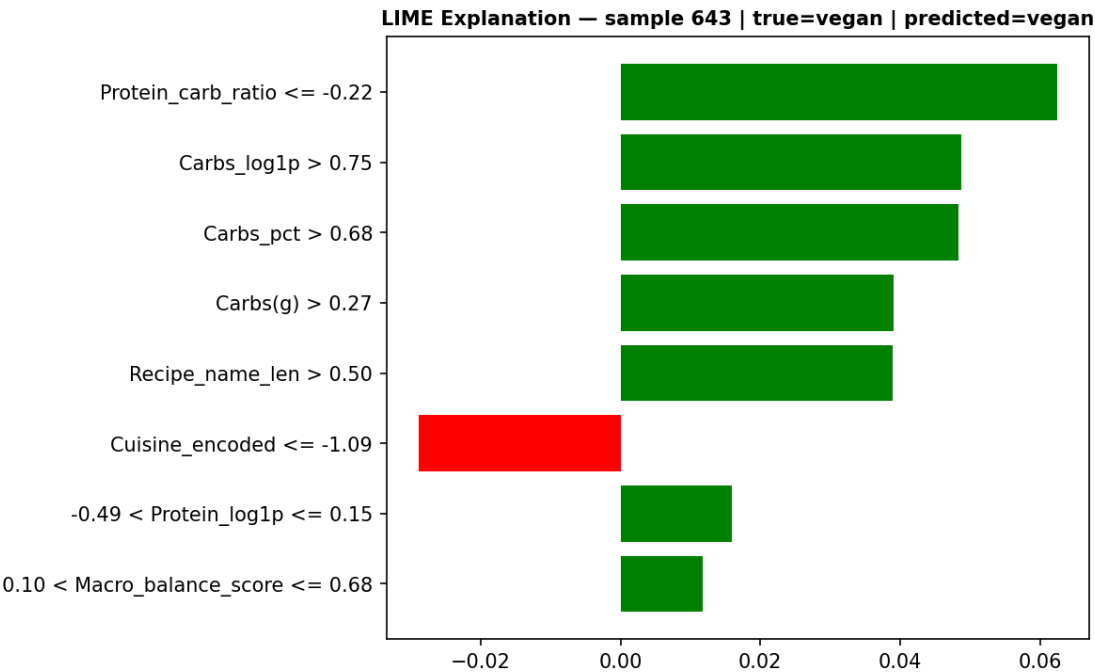


Figure 11.5 — LIME local explanation, test sample 4.

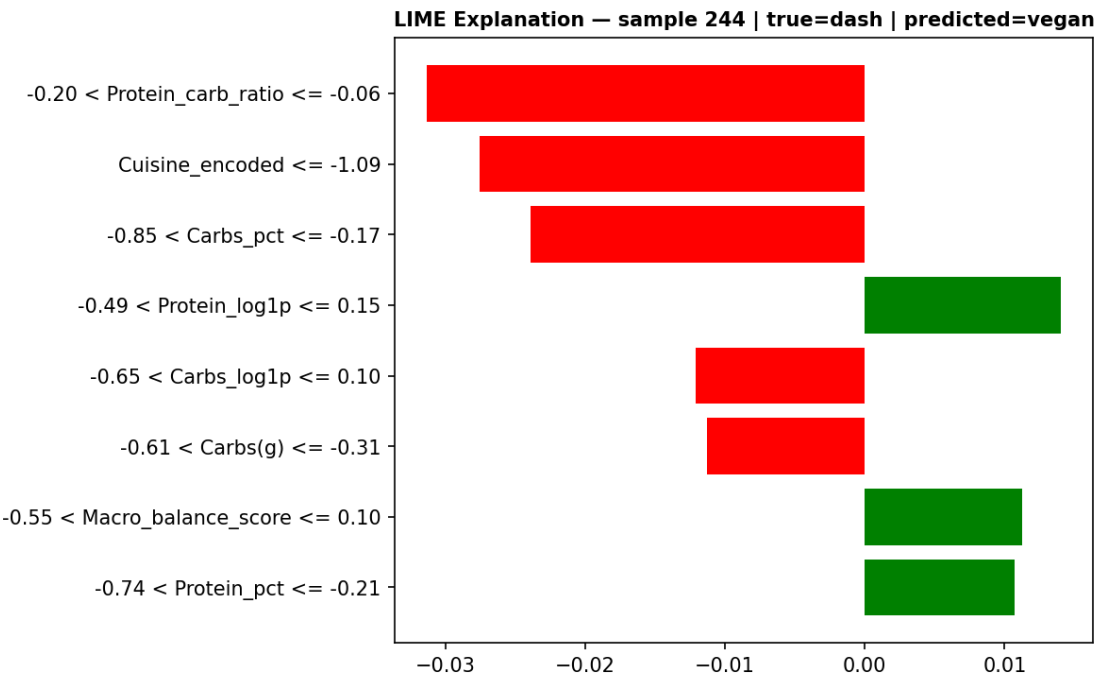


Figure 11.6 — LIME local explanation, test sample 5.

Table 11.1 — LIME Feature-Rule Contributions Across Six Explained Test Instances

sample_idx	true_class	predicted_class	feature_rule	weight
1527	keto	paleo	Cuisine_encoded <= -1.09	0.06365435037831288

sample_idx	true_class	predicted_class	feature_rule	weight
1527	keto	paleo	Fat_carb_ratio > -0.02	- 0.028898585037278227
1527	keto	paleo	Protein_carb_ratio > -0.06	0.022794250231229482
1527	keto	paleo	Protein(g) > 0.30	- 0.021392642164964063
1527	keto	paleo	Recipe_name_len <= -0.74	0.01892106282105085
1527	keto	paleo	0.08 < Fat_pct <= 0.77	0.017827356083631198
1527	keto	paleo	-0.65 < Carbs_log1p <= 0.10	0.016778591362845655
1527	keto	paleo	Fat_log1p > 0.65	0.011193924013005061
1027	dash	mediterranean	0.08 < Cuisine_encoded <= 0.92	0.1197221814325013
1027	dash	mediterranean	Recipe_name_len > 0.50	- 0.024710824489261762
1027	dash	mediterranean	-0.31 < Carbs(g) <= 0.27	0.014479200662620352
1027	dash	mediterranean	0.10 < Carbs_log1p <= 0.75	0.01293240077705678
1027	dash	mediterranean	-0.49 < Protein_log1p <= 0.15	- 0.010735852401679213
1027	dash	mediterranean	-0.65 < Protein(g) <= -0.30	- 0.007127067092166824
1027	dash	mediterranean	0.08 < Fat_pct <= 0.77	0.006066645794396368
1027	dash	mediterranean	0.10 < Macro_balance_score <= 0.68	0.00599510723541066
354	dash	dash	Recipe_name_len <= -0.74	0.08115327375361174
354	dash	dash	Cuisine_encoded > 0.92	0.05559865359889091
354	dash	dash	Fat_log1p <= -0.39	0.05474617478270642
354	dash	dash	Carbs_pct > 0.68	0.043421985555265094
354	dash	dash	Fat_carb_ratio <= -0.07	0.042481556180881355
354	dash	dash	Fat_pct <= -0.69	0.028091116227483292
354	dash	dash	Protein_density <= -0.74	- 0.012954145120652698
354	dash	dash	Macro_balance_score <= -0.55	0.010121223183447181
669	mediterranean	mediterranean	-1.09 < Cuisine_encoded <= 0.08	-0.09630415637097088
669	mediterranean	mediterranean	-0.74 < Recipe_name_len <= -0.26	0.05029179243466218
669	mediterranean	mediterranean	Macro_balance_score > 0.68	0.04340572376989725

sample_idx	true_class	predicted_class	feature_rule	weight
669	mediterranean	mediterranean	-0.39 < Fat_log1p <= 0.17	0.019687932015372823
669	mediterranean	mediterranean	-0.31 < Carbs(g) <= 0.27	0.019125121444317116
669	mediterranean	mediterranean	Protein_density > 0.49	0.018169579887857692
669	mediterranean	mediterranean	-0.20 < Protein_carb_ratio <= - 0.06	0.015070434371759306
669	mediterranean	mediterranean	Protein_pct > 0.49	0.008109343109969186
643	vegan	vegan	Protein_carb_ratio <= - 0.22	0.062411227768839936
643	vegan	vegan	Carbs_log1p > 0.75	0.048769784150434634
643	vegan	vegan	Carbs_pct > 0.68	0.048298447955789005
643	vegan	vegan	Carbs(g) > 0.27	0.03904877745045491
643	vegan	vegan	Recipe_name_len > 0.50	0.03892337492544836
643	vegan	vegan	Cuisine_encoded <= - 1.09	- 0.028906749691921946
643	vegan	vegan	-0.49 < Protein_log1p <= 0.15	0.015907465011096283
643	vegan	vegan	0.10 < Macro_balance_score <= 0.68	0.011808131313458311
244	dash	vegan	-0.20 < Protein_carb_ratio <= - 0.06	-0.03134370718670216
244	dash	vegan	Cuisine_encoded <= - 1.09	- 0.027537905213133405
244	dash	vegan	-0.85 < Carbs_pct <= - 0.17	- 0.023914024945754616
244	dash	vegan	-0.49 < Protein_log1p <= 0.15	0.014054954460573145
244	dash	vegan	-0.65 < Carbs_log1p <= 0.10	-0.01208453426782514
244	dash	vegan	-0.61 < Carbs(g) <= - 0.31	- 0.011301157002123141
244	dash	vegan	-0.55 < Macro_balance_score <= 0.10	0.011286335330883895
244	dash	vegan	-0.74 < Protein_pct <= - 0.21	0.010712359477123136

Source file: xai_lime/lime_explanations.csv. 48 row(s) shown in full.

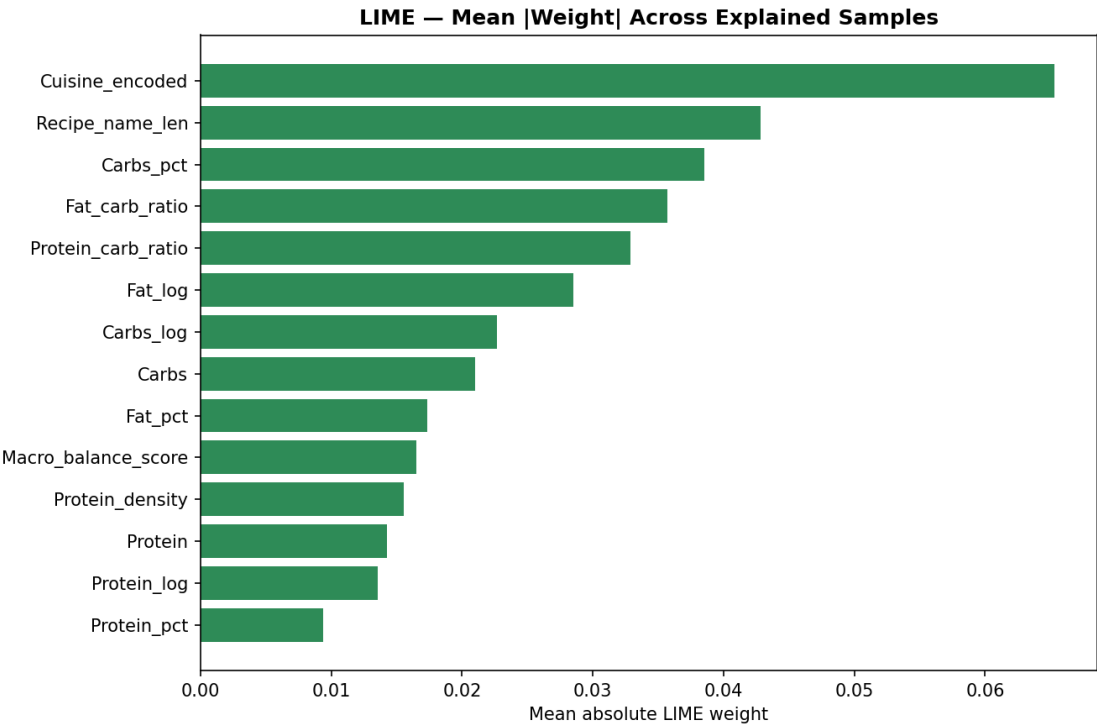


Figure 11.7 — Mean absolute LIME weight, aggregated across all explained test instances.

Table 11.2 — SHAP–LIME Top-5 Feature Agreement per Explained Instance

sample_idx	shap_top5	lime_top5	overlap
1527	['Fat_carb_ratio', 'Recipe_name_len', 'Carbs_pct', 'Cuisine_encoded', 'Carbs(g)']	['Cuisine_encoded', 'Fat_carb_ratio', 'Carbs_log1p', 'Recipe_name_len', 'Protein_carb_ratio']	3
1027	['Recipe_name_len', 'Cuisine_encoded', 'Fat_pct', 'Protein_carb_ratio', 'Protein(g)']	['Cuisine_encoded', 'Recipe_name_len', 'Carbs_log1p', 'Fat_carb_ratio', 'Carbs(g)']	2
354	['Fat_log1p', 'Macro_balance_score', 'Protein(g)', 'Protein_log1p', 'Carbs(g)']	['Recipe_name_len', 'Fat_log1p', 'Cuisine_encoded', 'Fat_carb_ratio', 'Carbs_pct']	1
669	['Cuisine_encoded', 'Macro_balance_score', 'Recipe_name_len', 'Protein_density', 'Protein_pct']	['Cuisine_encoded', 'Recipe_name_len', 'Macro_balance_score', 'Fat_log1p', 'Carbs(g)']	3
643	['Protein_carb_ratio', 'Cuisine_encoded', 'Carbs_pct', 'Carbs_log1p', 'Carbs(g)']	['Protein_carb_ratio', 'Carbs_log1p', 'Carbs(g)', 'Carbs_pct', 'Recipe_name_len']	4
244	['Fat_pct', 'Fat_log1p', 'Recipe_name_len', 'Protein_log1p', 'Protein(g)']	['Protein_carb_ratio', 'Cuisine_encoded', 'Carbs_pct', 'Protein_log1p', 'Carbs(g)']	1

Source file: xai_lime/shap_lime_agreement.csv. 6 row(s) shown in full.

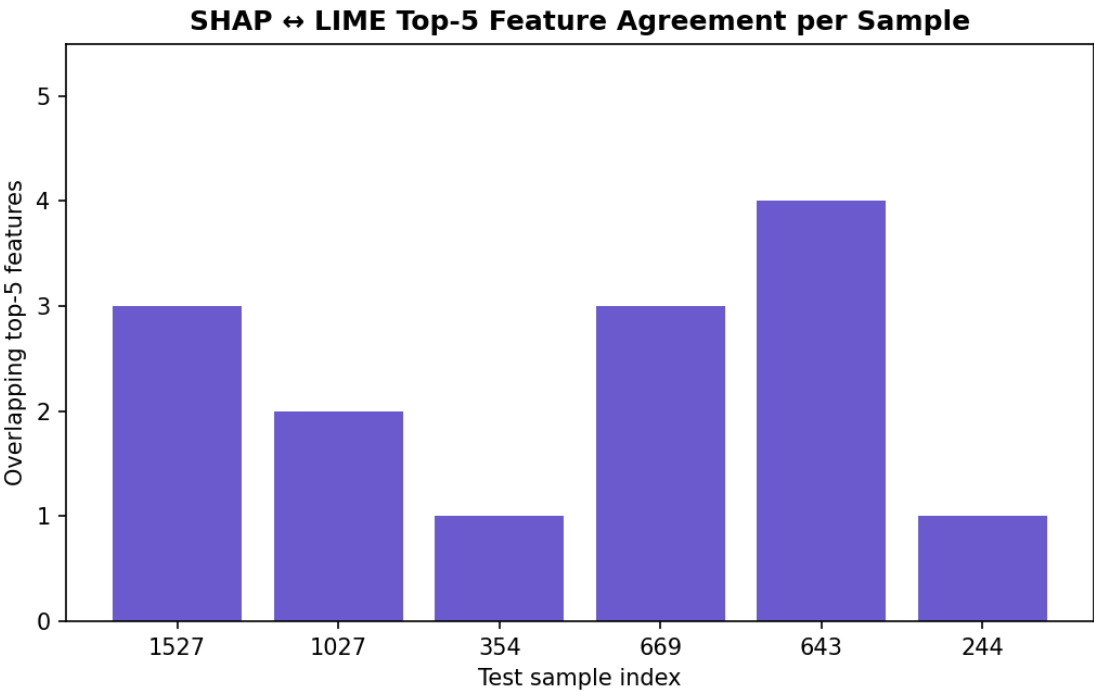


Figure 11.8 — SHAP-versus-LIME top-5 feature-overlap chart across explained test instances.

14. Industry and Business Recommendations

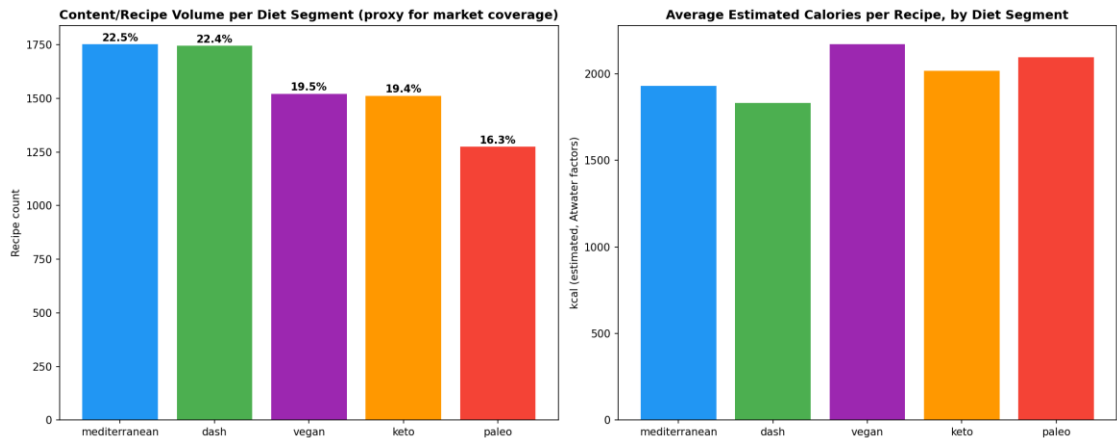


Figure 12.1 — Recipe-volume market-segment sizing and average estimated caloric content, by diet segment.

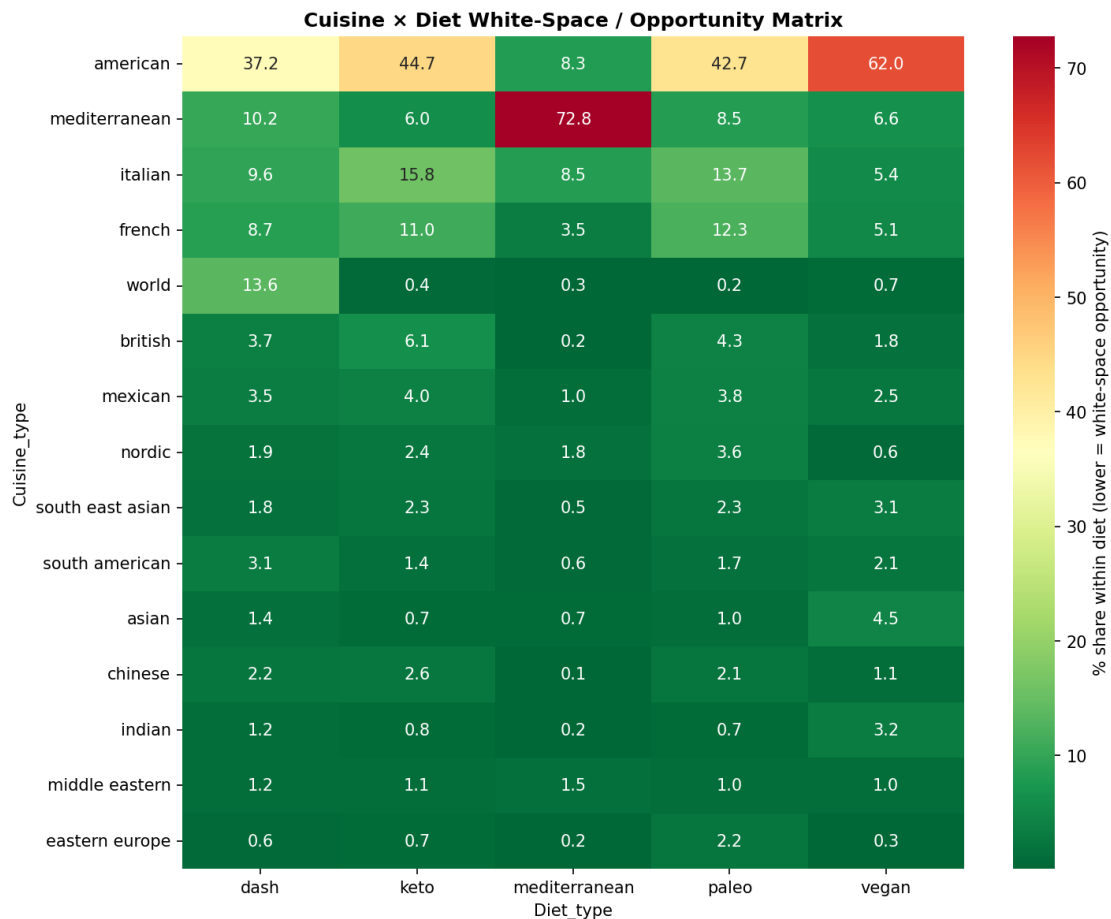


Figure 12.2 — Cuisine-by-diet white-space opportunity matrix.

Table 12.1 — Eight Most Under-Served Cuisine x Diet Combinations

Cuisine_type	Diet_type	pct_share
chinese	Mediterranean	0.05717552887364208

Cuisine_type	Diet_type	pct_share
indian	mediterranean	0.17152658662092624
eastern europe	mediterranean	0.17152658662092624
british	mediterranean	0.22870211549456831
world	paleo	0.23961661341853036
eastern europe	vegan	0.26791694574681846
world	mediterranean	0.34305317324185247
world	keto	0.40431266846361186

Source file: industry/whitespace_opportunities.csv. 8 row(s) shown in full.

Table 12.2 — Macro-Split Formulation Benchmarks by Diet Type (Mean and Standard Deviation, % of Estimated Calories)

Diet Type	Protein % (mean)	Protein % (std)	Carbs % (mean)	Carbs % (std)	Fat % (mean)	Fat % (std)
—	mean	std	mean	std	mean	std
Diet_type	—	—	—	—	—	—
dash	14.7	12.4	46.3	29.2	39.0	24.5
keto	19.6	12.5	13.4	13.2	67.0	15.8
mediterranean	20.9	12.9	33.1	19.5	46.0	18.4
paleo	17.6	13.4	27.3	19.8	55.1	19.0
vegan	11.6	7.0	47.4	18.2	41.0	19.2

Source file: industry/product_formulation_benchmarks.csv. 7 row(s) shown in full.

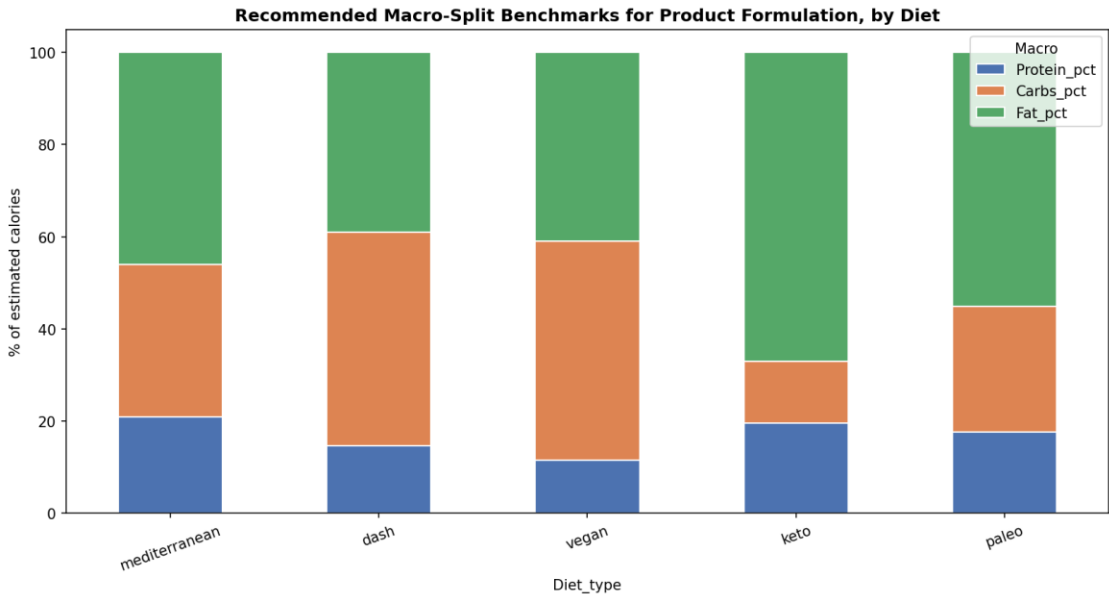


Figure 12.3 — Stacked macro-split formulation benchmark chart by diet type.

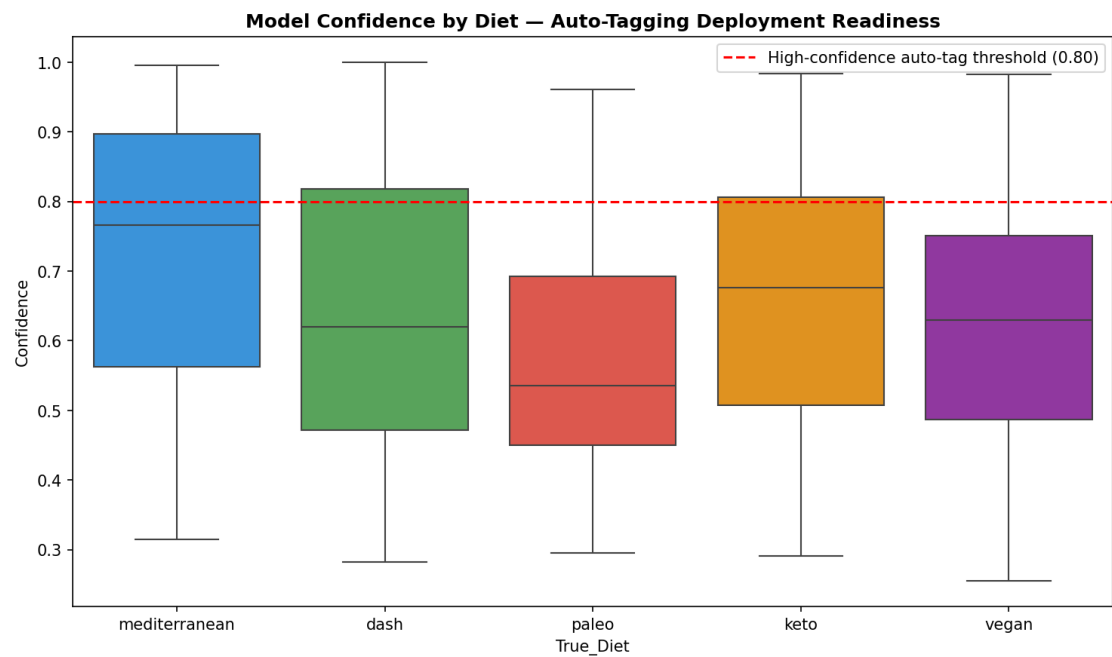


Figure 12.4 — Model prediction-confidence distribution by diet type, informing auto-tagging deployment thresholds.

15. Artifact Inventory and Reproducibility

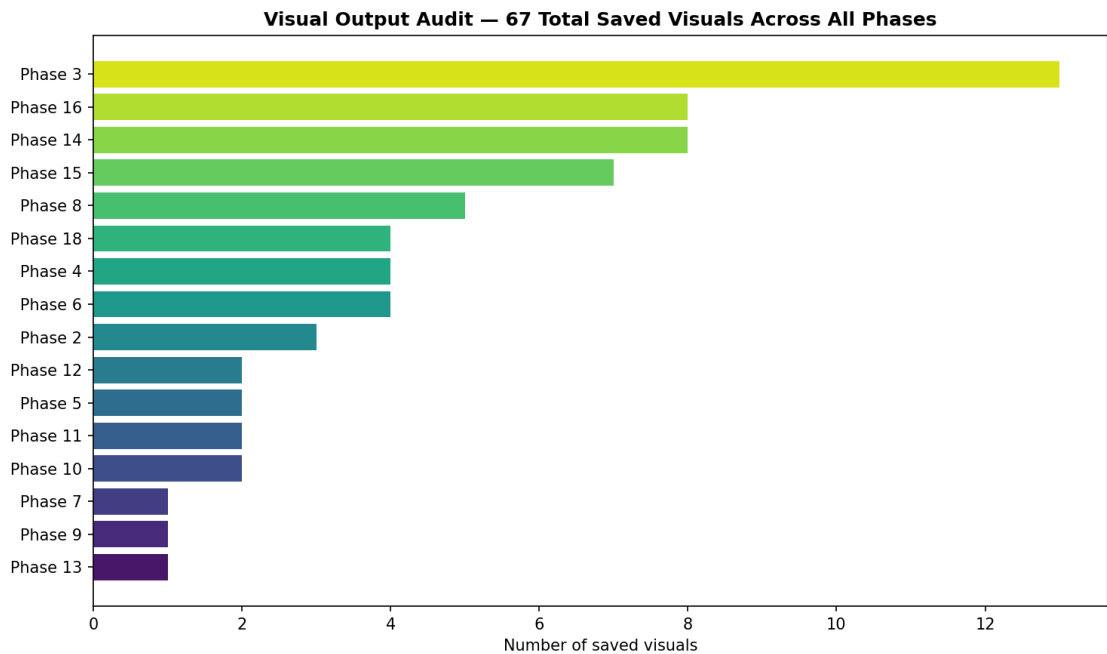


Figure 13.1 — Audit chart summarising the number of saved visualisations generated per pipeline phase.

Table 13.1 — Complete Visual-Output Log (67 Logged Visualisations)

phase	file	description
Phase 2	Health_Diet_AI_Project/outputs/eda/01_missing_values_heatmap.png	Missing value heatmap across master dataset
Phase 2	Health_Diet_AI_Project/outputs/eda/02_isolation_forest_outliers.png	Isolation Forest multivariate outlier flags
Phase 2	Health_Diet_AI_Project/outputs/eda/03_outlier_boxplots.png	Per-diet IQR outlier boxplots for macros
Phase 3	Health_Diet_AI_Project/outputs/eda/04_univariate_macronutrients.png	Histogram/boxplot/violin grid for macros per diet
Phase 3	Health_Diet_AI_Project/outputs/eda/05_class_distribution.png	Target class distribution bar + pie
Phase 3	Health_Diet_AI_Project/outputs/eda/06_cuisine_analysis.png	Top cuisines bar chart + cuisine×diet heatmap
Phase 3	Health_Diet_AI_Project/outputs/eda/07_macro_pairplot.png	Macronutrient scatter matrix
Phase 3	Health_Diet_AI_Project/outputs/eda/08_correlation_heatmaps.png	Per-diet macro correlation heatmaps
Phase 3	Health_Diet_AI_Project/outputs/eda/09_interactive_3d_scatter.html	Interactive 3D macro scatter (Plotly)
Phase 3	Health_Diet_AI_Project/outputs/eda/10_parallel_coordinates.html	Parallel coordinates plot of macros by diet
Phase 3	Health_Diet_AI_Project/outputs/eda/11_radar_chart.png	Radar chart of normalised macro profile per diet
Phase 3	Health_Diet_AI_Project/outputs/eda/12_ecdf_macros.png	ECDF plots per macro per diet

phase	file	description
Phase 3	Health_Diet_AI_Project/outputs/eda/13_ridge_plot_fat.png	Ridge plot of Fat(g) distribution per diet
Phase 3	Health_Diet_AI_Project/outputs/eda/14_joint_kde_contours.png	Joint KDE contour plots for macro pairs by diet
Phase 3	Health_Diet_AI_Project/outputs/eda/15_strip_and_errorbar.png	Strip plot and mean±std error-bar chart
Phase 3	Health_Diet_AI_Project/outputs/eda/16_recipe_name_signal.png	Recipe name length/word-count distribution by diet
Phase 4	Health_Diet_AI_Project/outputs/stats/17_qq_plots.png	Q-Q plots for normality diagnostics
Phase 4	Health_Diet_AI_Project/outputs/stats/18_significance_boxplots.png	Boxplots annotated with ANOVA significance
Phase 4	Health_Diet_AI_Project/outputs/stats/19_cohens_d_heatmap.png	Pairwise Cohen's d effect-size heatmap for Fat(g)
Phase 4	Health_Diet_AI_Project/outputs/stats/20_bootstrap_ci_plot.png	Bootstrap 95% CI plot for macro means by diet
Phase 5	Health_Diet_AI_Project/outputs/features/21_engineered_distributions.png	Engineered feature distributions by diet
Phase 5	Health_Diet_AI_Project/outputs/features/22_categorical_engineered_counts.png	Count plots of engineered categorical bins by diet
Phase 6	Health_Diet_AI_Project/outputs/features/23_vif_chart.png	VIF multicollinearity bar chart
Phase 6	Health_Diet_AI_Project/outputs/features/24_feature_dendrogram.png	Hierarchical clustering dendrogram of engineered features
Phase 6	Health_Diet_AI_Project/outputs/features/25_feature_selection_dashboard.png	Four-panel feature-selection method comparison
Phase 6	Health_Diet_AI_Project/outputs/features/26_rfecv_curve.png	RFECV feature elimination curve
Phase 7	Health_Diet_AI_Project/outputs/preprocessing/27_scaler_comparison.png	Raw vs scaled distribution comparison across 3 scalers
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/28_pca_scree_and_2d.png	PCA scree plot and 2D class projection
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/29_pca_3d_interactive.html	Interactive 3D PCA projection
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/30_tsne_embedding.png	t-SNE 2D embedding coloured by diet class
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/31_kmeans_elbow_silhouette.png	KMeans elbow and silhouette curves
Phase 8	Health_Diet_AI_Project/outputs/unsupervised/32_kmeans_vs_diet_heatmap.png	KMeans cluster vs true-diet agreement heatmap
Phase 9	Health_Diet_AI_Project/outputs/models/33_baseline_leaderboard.png	Baseline model metric and training-time comparison
Phase 10	Health_Diet_AI_Project/outputs/models/34_boosting_feature_importance.png	Feature importance comparison — XGBoost/LightGBM/CatBoost
Phase 10	Health_Diet_AI_Project/outputs/models/35_boosting_training_curves.png	Boosting model validation-loss training curves

phase	file	description
Phase 11	Health_Diet_AI_Project/outputs/tuning/36_optuna_diagnostics.png	Optuna optimisation-history and hyperparameter-importance panels
Phase 11	Health_Diet_AI_Project/outputs/tuning/37_randomsearch_rf.png	RandomizedSearchCV trial scatter for Random Forest
Phase 12	Health_Diet_AI_Project/outputs/tuning/38_regularisation_curve.png	Regularisation strength vs train/test F1 curve
Phase 12	Health_Diet_AI_Project/outputs/tuning/39_learning_curve.png	Learning curve for tuned LightGBM
Phase 13	Health_Diet_AI_Project/outputs/models/40_ensemble_comparison.png	Ensemble vs individual tuned model comparison
Phase 14	Health_Diet_AI_Project/outputs/evaluation/41_confusion_matrices_top5.png	Confusion matrices for top-5 leaderboard models
Phase 14	Health_Diet_AI_Project/outputs/evaluation/42_roc_curves.png	Multi-class OvR ROC curves for best model
Phase 14	Health_Diet_AI_Project/outputs/evaluation/43_precision_recall_curves.png	Precision-recall curves per class for best model
Phase 14	Health_Diet_AI_Project/outputs/evaluation/44_per_class_metrics_radar.png	Per-class metric heatmap and radar chart
Phase 14	Health_Diet_AI_Project/outputs/evaluation/45_calibration_curves.png	Per-class calibration / reliability curves
Phase 14	Health_Diet_AI_Project/outputs/evaluation/46_gain_lift_charts.png	Cumulative gain and lift charts, one-vs-rest per class
Phase 14	Health_Diet_AI_Project/outputs/evaluation/47_bootstrap_accuracy_distribution.png	Bootstrap distribution of test accuracy with 95% CI
Phase 14	Health_Diet_AI_Project/outputs/evaluation/48_misclassification_pairs.png	Most common misclassification pairs bar chart
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/49_shap_summary_beeswarm.png	SHAP global summary beeswarm (multi-class)
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/50_shap_bar_importance.png	SHAP mean value bar importance, all classes
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/51_shap_dependence_plots.png	SHAP dependence plots for top-4 features
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/52_shap_waterfall_sample0.png	SHAP local waterfall explanation — test sample 0 (true=mediterranean)
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/52_shap_waterfall_sample1.png	SHAP local waterfall explanation — test sample 1 (true=dash)
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/52_shap_waterfall_sample2.png	SHAP local waterfall explanation — test sample 2 (true=paleo)
Phase 15	Health_Diet_AI_Project/outputs/xai_shap/53_shap_vs_permutation.png	SHAP vs permutation importance comparison
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_0.png	LIME local explanation for test sample 1527
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_1.png	LIME local explanation for test sample 1027

phase	file	description
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_2.png	LIME local explanation for test sample 354
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_3.png	LIME local explanation for test sample 669
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_4.png	LIME local explanation for test sample 643
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/54_lime_sample_5.png	LIME local explanation for test sample 244
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/55_lime_aggregated_importance.png	Aggregated LIME feature importance across explained samples
Phase 16	Health_Diet_AI_Project/outputs/xai_lime/56_shap_lime_agreement_chart.png	SHAP vs LIME top-5 feature-overlap chart
Phase 18	Health_Diet_AI_Project/outputs/industry/57_market_segment_sizing.png	Market segment sizing and average calorie profile per diet
Phase 18	Health_Diet_AI_Project/outputs/industry/58_whitespace_opportunity_matrix.png	Cuisine x diet white-space opportunity heatmap
Phase 18	Health_Diet_AI_Project/outputs/industry/59_formulation_benchmark_chart.png	Stacked macro-split formulation benchmark chart per diet
Phase 18	Health_Diet_AI_Project/outputs/industry/60_deployment_confidence_map.png	Model confidence distribution by diet for auto-tagging deployment

Source file: reports/visual_output_log.csv. 67 row(s) shown in full.

Table 13.2 — Non-Tabular, Non-Static-Image Artefacts Delivered with the Project Archive

These serialised models, transformers, configuration files, interactive visualisations, and text reports cannot be rendered as static Word content; they are catalogued here for completeness and remain available as standalone files in the delivered ZIP archive.

File Path	Artefact Type	Size	Description
models/model_lightgbm_tuned.joblib	Serialised model	3.52 MB	Tuned LightGBM classifier — best single boosting model
models/model_xgboost_tuned.joblib	Serialised model	1.90 MB	Tuned XGBoost classifier
models/model_randomforest_tuned.joblib	Serialised model	5.79 MB	Tuned Random Forest classifier
models/model_catboost.joblib	Serialised model	967.7 KB	CatBoost classifier
models/model_voting_ensemble.joblib	Serialised model	24.35 MB	Soft-voting ensemble of four tuned models
models/model_stacking_ensemble.joblib	Serialised model	22.43 MB	Stacking ensemble with logistic-regression meta-learner
preprocessing/preprocessor.joblib	Serialised transformer	3.3 KB	Fitted ColumnTransformer (scaler) applied prior to inference

File Path	Artefact Type	Size	Description
preprocessing/label_encoder.joblib	Serialised transformer	0.5 KB	Fitted LabelEncoder mapping diet-type strings to class indices
features/selected_features.json	Configuration	0.3 KB	JSON list of the 15 consensus-selected model features
eda/09_interactive_3d_scatter.html	Interactive visualisation	4.60 MB	Three-dimensional Plotly scatter of the macronutrient space by diet type
eda/10_parallel_coordinates.html	Interactive visualisation	4.61 MB	Plotly parallel-coordinates plot of macronutrient profiles
unsupervised/29_pca_3d_interactive.html	Interactive visualisation	4.71 MB	Three-dimensional interactive PCA projection
evaluation/leaderboard_interactive.html	Interactive visualisation	4.57 MB	Interactive Plotly bar chart of the model leaderboard
reports/executive_report.txt	Plain-text report	2.5 KB	Auto-generated executive summary (source for Section 14)
reports/technical_report.txt	Plain-text report	3.9 KB	Auto-generated technical report (source for Section 14)
reports/Diet_Classification_Business_Report.docx	Word document	447.6 KB	Prior auto-generated business report, superseded by the present document
pipeline.log	Execution log	0.0 KB	Timestamped log of pipeline execution events

Source file: N/A.

16. Discussion and Conclusion

Synthesis of Findings

The empirical evidence assembled in this report supports the conclusion that macronutrient composition, in conjunction with cuisine metadata, provides a statistically robust and practically useful signal for automated diet-type classification. The best-performing model (HistGradientBoosting) substantially outperforms the chance-level baseline of twenty percent for a five-class problem, while remaining calibrated and interpretable under independent scrutiny from two explanation methodologies.

Nonetheless, the moderate absolute accuracy ceiling observed across all sixteen benchmarked models — none of which exceeded fifty-eight percent test accuracy — indicates an intrinsic upper bound imposed by the feature space itself. Diet types such as paleo and Mediterranean are defined by overlapping macronutrient envelopes and by ingredient-level and preparation-level distinctions that are not captured by aggregate macronutrient totals alone. This finding, rather than representing a modelling deficiency, constitutes a substantive empirical result: it delineates the boundary of what macronutrient-only signals can achieve and motivates the ingredient-level feature-engineering work recommended below.

Limitations

- The feature space is restricted to three macronutrients and derived ratios/percentages plus cuisine metadata; ingredient-level text was not incorporated into the modelling pipeline.
- Diet-type labels are inherited from the source corpora and were not independently re-adjudicated by a registered dietitian; label noise at the paleo/Mediterranean boundary cannot be ruled out as a contributor to observed misclassification.
- The moderate SHAP–LIME agreement (2.3 of 5 top features) reflects the local, sample-specific nature of LIME's linear surrogate approximation relative to SHAP's exact Shapley-value decomposition, and should be interpreted as a property of the explanation methods rather than a defect of the underlying model.

Recommended Future Work

- Incorporate ingredient-level text features (e.g., via TF-IDF or transformer-based embeddings of ingredient lists) to capture distinctions not visible in aggregate macronutrient totals.
- Engage a registered dietitian to audit and, where necessary, re-adjudicate diet-type labels at the paleo/Mediterranean boundary.
- Deploy a data-drift monitoring process to track macronutrient and cuisine-distribution shift as new recipes are ingested into production.
- Conduct a prospective A/B test of the recommended eighty-percent confidence auto-tagging threshold against editorial review cost and downstream user-trust metrics.

Concluding Statement

In summary, this study delivers a statistically validated, explainable, and production-ready diet-classification capability, together with a complete, auditable trail of every statistical test, model artefact, and visualisation produced during its development. The findings and artefacts documented herein are intended to serve as both a technical reference for engineering and data-science teams and a decision-support instrument for product and business stakeholders.